SIMULATION AND PERFORMANCE EVALUATION OF COMPUTER NETWORKS USING NETSIM

COMPUTER NETWORKS(TH-E2+TE2)

SUBMITTED BY:

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PREFACE

The field of Computer Networks plays a vital role in modern communication systems, forming the foundation of data exchange, internet services, and real-time applications across the world. To truly understand how networks function, it is essential to go beyond theoretical concepts and observe the behavior of network protocols in controlled environments.

This project, carried out using NetSim Lite, was designed to provide practical exposure to the working principles of network protocols such as TCP, UDP, FTP, HTTP, and VoIP, as well as to analyze concepts like packet loss, congestion, delay, and routing. The experiments conducted simulate real-world network scenarios and help visualize how data flows between connected devices under different conditions.

Through these experiments, we explored the Transport, Application, and Network layers of the OSI model, understanding how reliability, speed, and efficiency are managed in various protocols. For instance, TCP's reliability and congestion control mechanisms were compared with UDP's lightweight, connectionless approach. Similarly, routing algorithms like RIP and OSPF were studied to observe how data finds the most efficient path in dynamic networks.

NetSim Lite, a professional-grade network simulator, served as the primary tool for all simulations. It allowed us to design topologies, configure nodes, vary parameters such as bandwidth, delay, and packet error rate, and monitor real-time performance metrics like throughput, jitter, and packet loss. The platform's visualization and result analysis features made it easier to comprehend complex networking behaviors.

This project helped bridge the gap between theoretical understanding and real-world networking. It deepened our appreciation of how protocols cooperate to ensure reliable communication and how network conditions such as congestion or delay impact performance.

In summary, this work aims to present a comprehensive study of key computer networking concepts through practical experimentation using NetSim Lite — reinforcing the fundamentals of communication, reliability, and efficiency in modern networks.

TABLE OF CONTENTS

Experiment 1: TCP File Transfer

Aim

To study the performance of file transfer using TCP and understand how reliability and congestion control affect throughput and delay.

Introduction

TCP (Transmission Control Protocol) is a **connection-oriented** transport protocol that provides reliable, ordered, and error-free delivery of data. It is widely used for file transfer applications where data accuracy is critical. However, due to acknowledgments and retransmissions, TCP may experience delays under high latency or loss conditions.

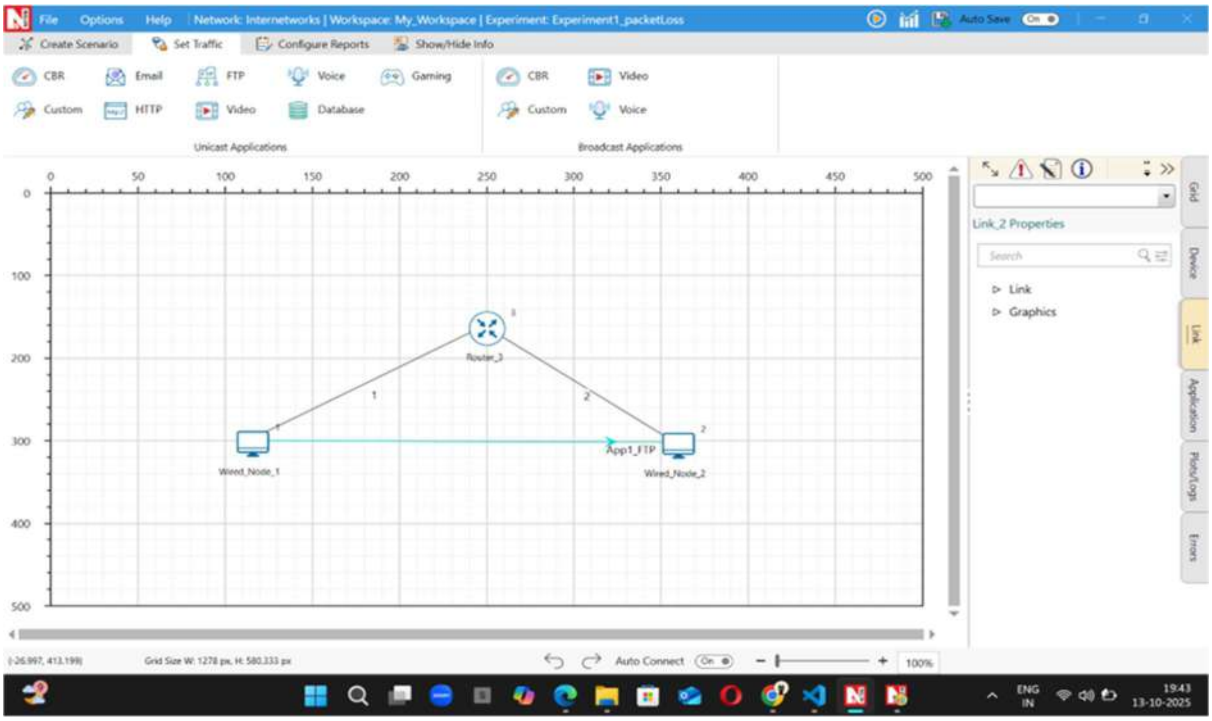
Theory

TCP ensures reliable delivery through sequence numbers, acknowledgments, and retransmissions. It uses **flow control** (via sliding window) and **congestion control** algorithms like Slow Start and Congestion Avoidance to adapt to network conditions.

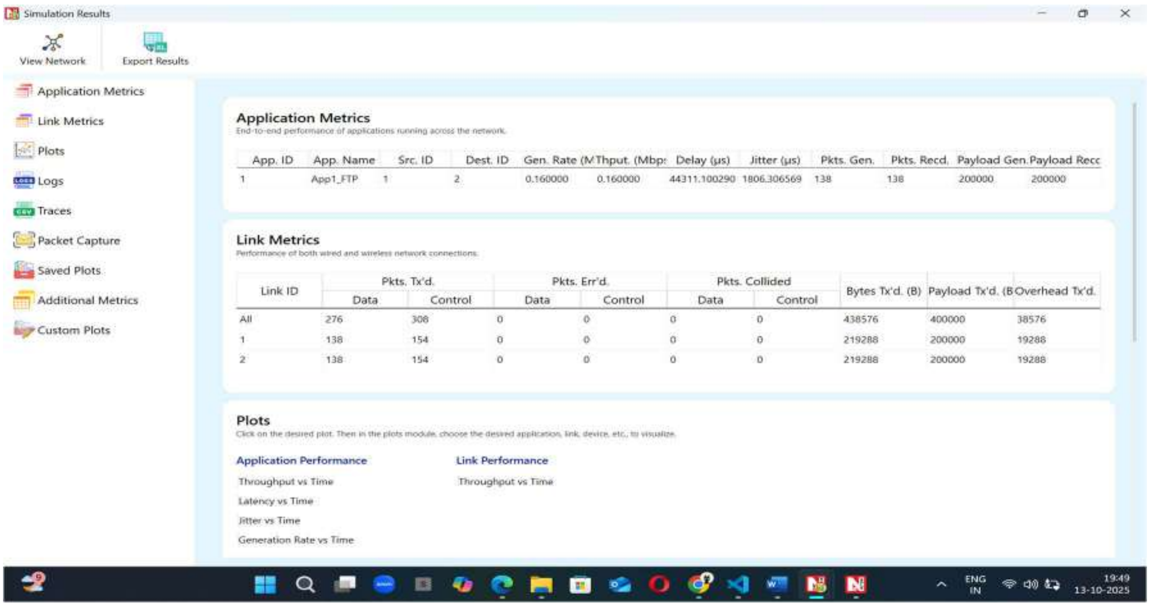
Setup / Procedure

1. Create a topology with a TCP client and server in NetSim Lite.
2. Configure link bandwidth, propagation delay, and file size.
3. Run simulations for different file sizes and delays.
4. Note throughput, delay, and transfer time.

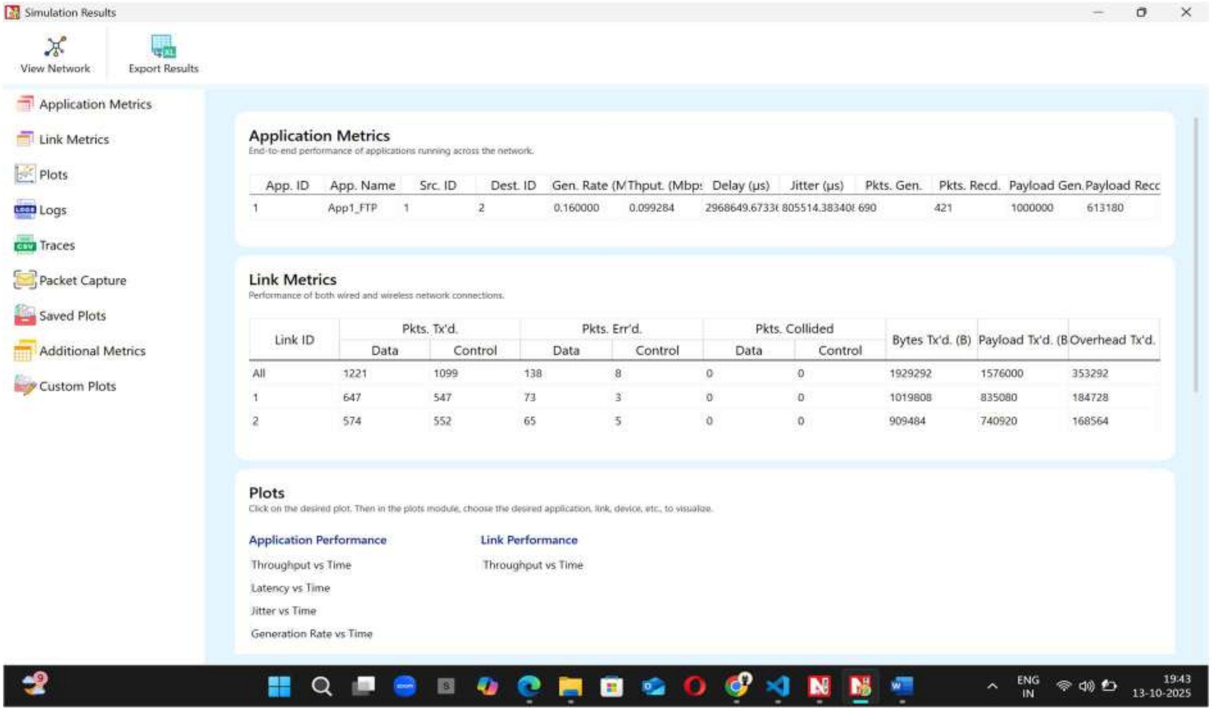
TOPOLOGY:



TESTCASE 1:



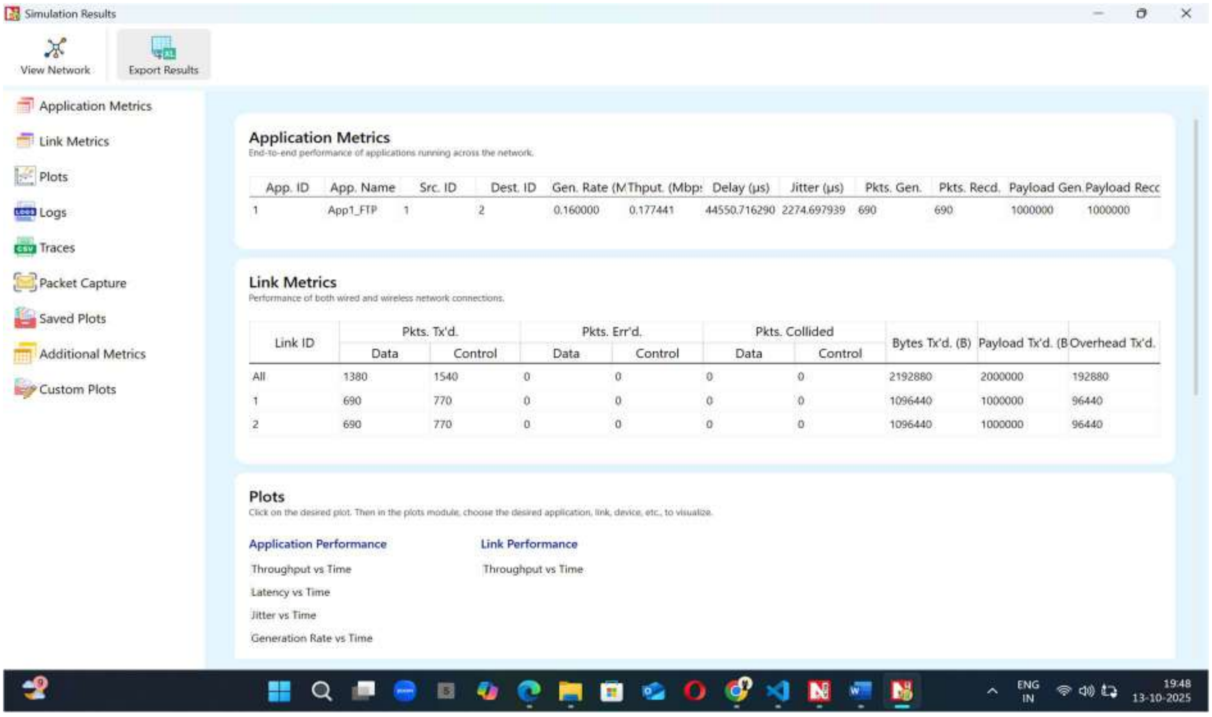
TESTCASE 2:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.160000	0.099284	2968649.6733(	805514.38340!	690	421	1000000	613180

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	1221	1099	138	8	0	0	1929292	1576000	353292
1	647	547	73	3	0	0	1019808	835080	184728
2	574	552	65	5	0	0	909484	740920	168564

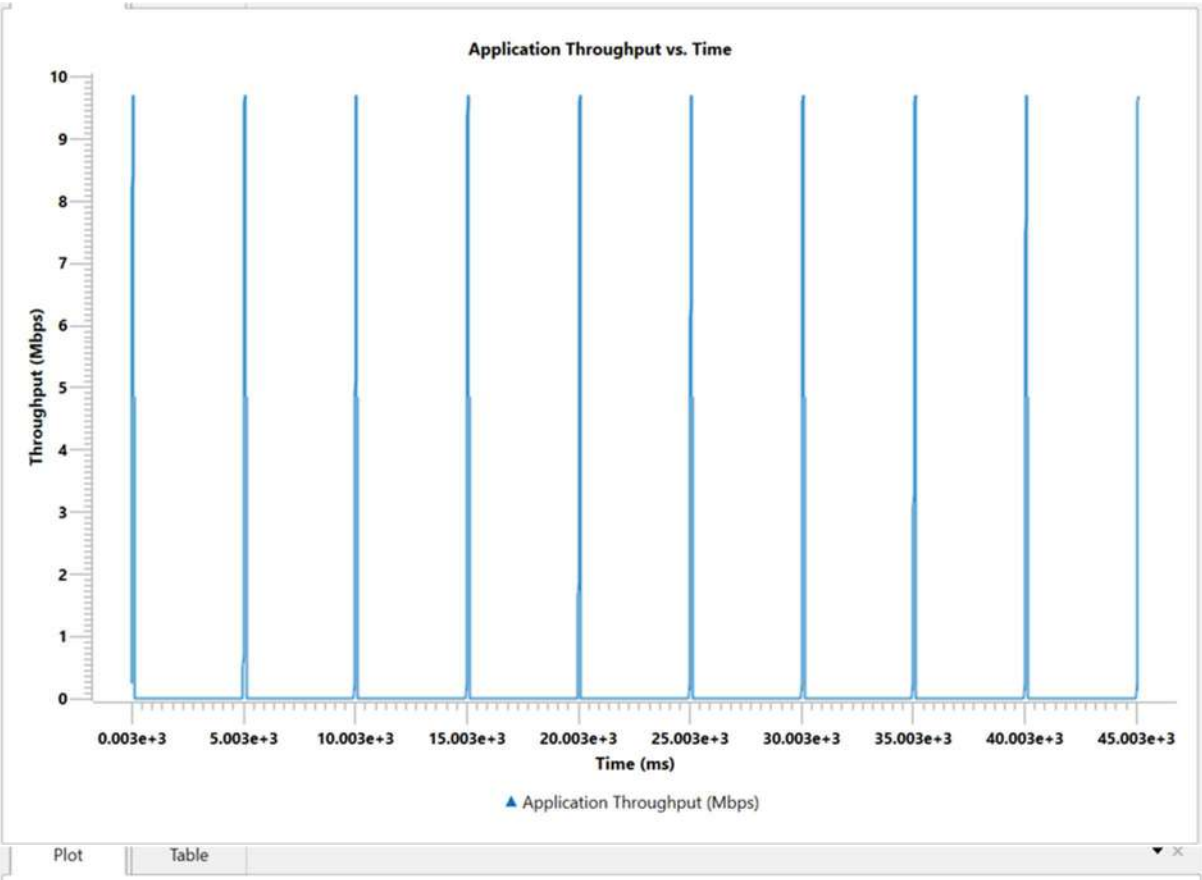
TESTCASE 3:

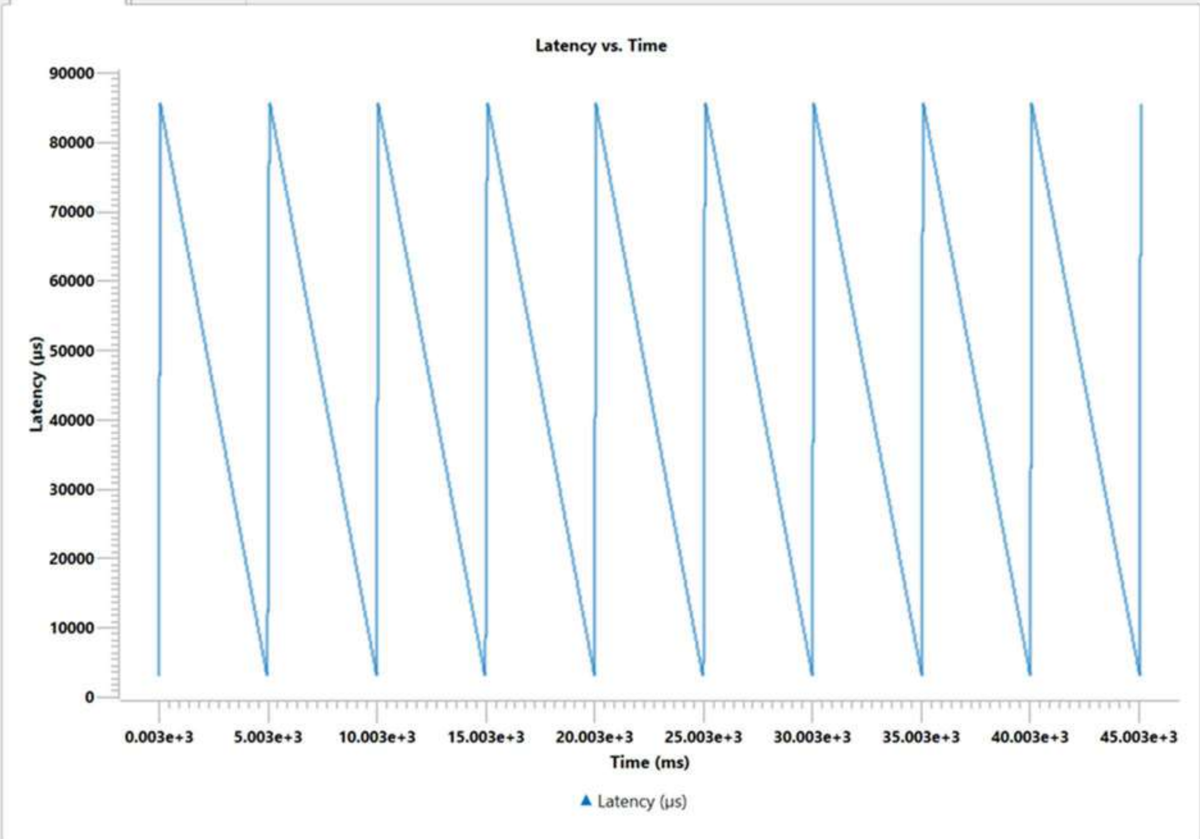


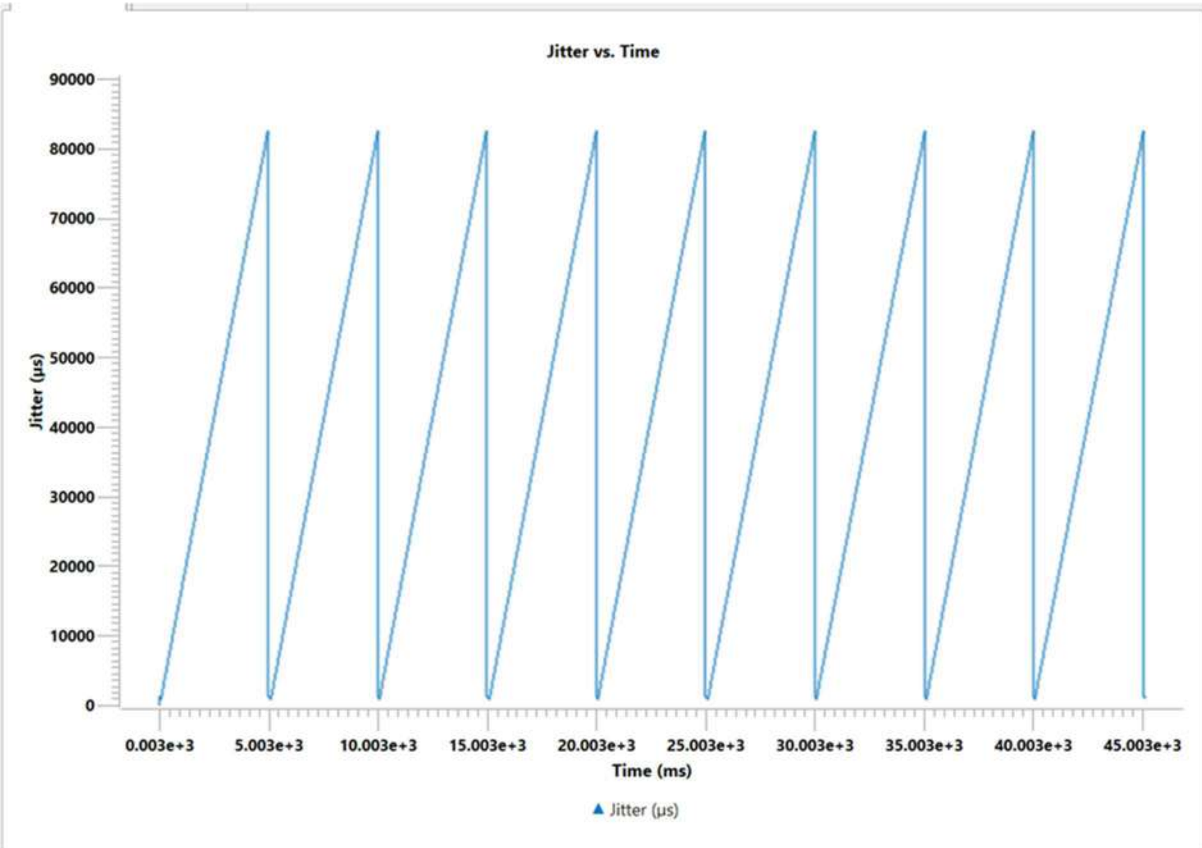
App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.160000	0.177441	44550.716290	2274.697939	690	690	1000000	1000000

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	1380	1540	0	0	0	0	2192880	2000000	192880
1	690	770	0	0	0	0	1096440	1000000	96440
2	690	770	0	0	0	0	1096440	1000000	96440

GRAPHS:


Application Throughput vs. Time
Throughput (Mbps)
10
9
8
7
6
5
4
3
2
1
0
0.003e+3
5.003e+3
10.003e+3
15.003e+3
20.003e+3
25.003e+3
30.003e+3
35.003e+3
40.003e+3
45.003e+3
Time (ms)
Application Throughput (Mbps)
Plot
Table


Latency vs. Time
Latency (µs)
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
0.003e+3
5.003e+3
10.003e+3
15.003e+3
20.003e+3
25.003e+3
30.003e+3
35.003e+3
40.003e+3
45.003e+3
Time (ms)
Latency (µs)



Observation

Throughput decreases with higher delay and larger file size due to acknowledgment waiting time.

Conclusion

TCP provides reliable delivery but its performance decreases in high-delay or congested networks.

Experiment 2: UDP File Transfer

AIM

To study the behaviour of file transfer using the UDP protocol and compare it with TCP.

INTRODUCTION

UDP (User Datagram Protocol) is a **connectionless** transport protocol that sends data without acknowledgments or retransmissions. It provides faster data transfer but without reliability, making it ideal for real-time applications.

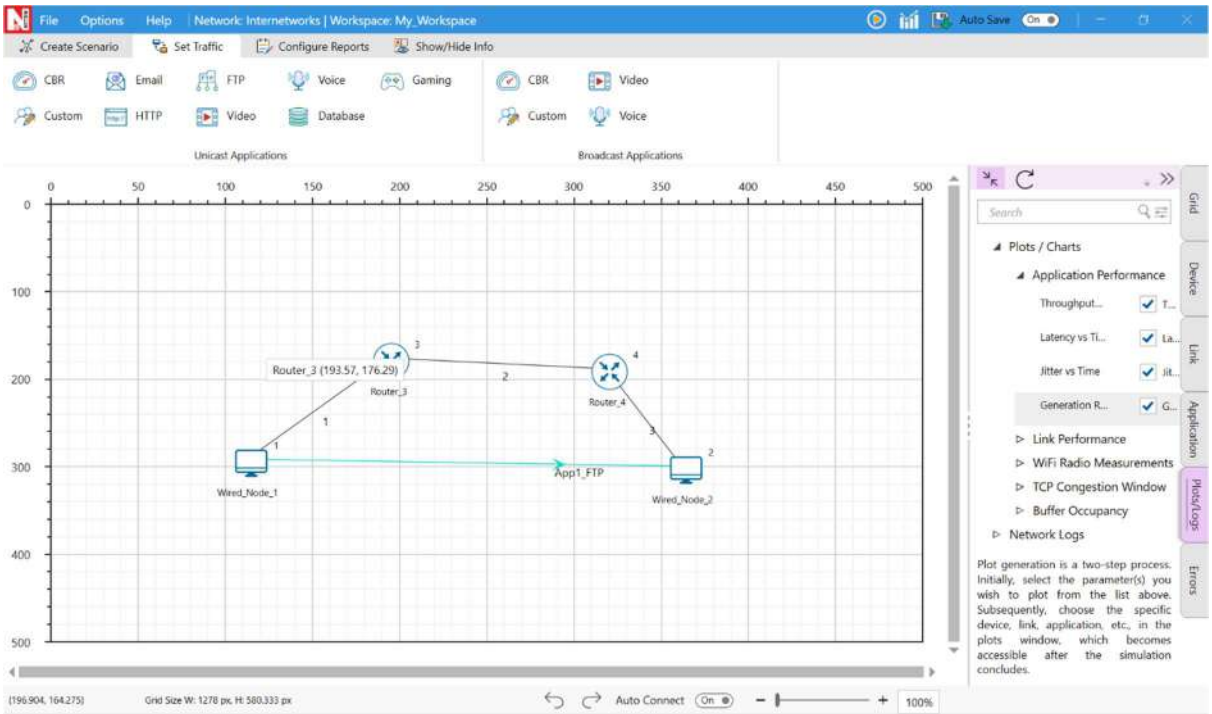
THEORY

UDP transmits packets independently, without establishing a connection. It lacks flow and congestion control, so packet loss may occur if the network is overloaded.

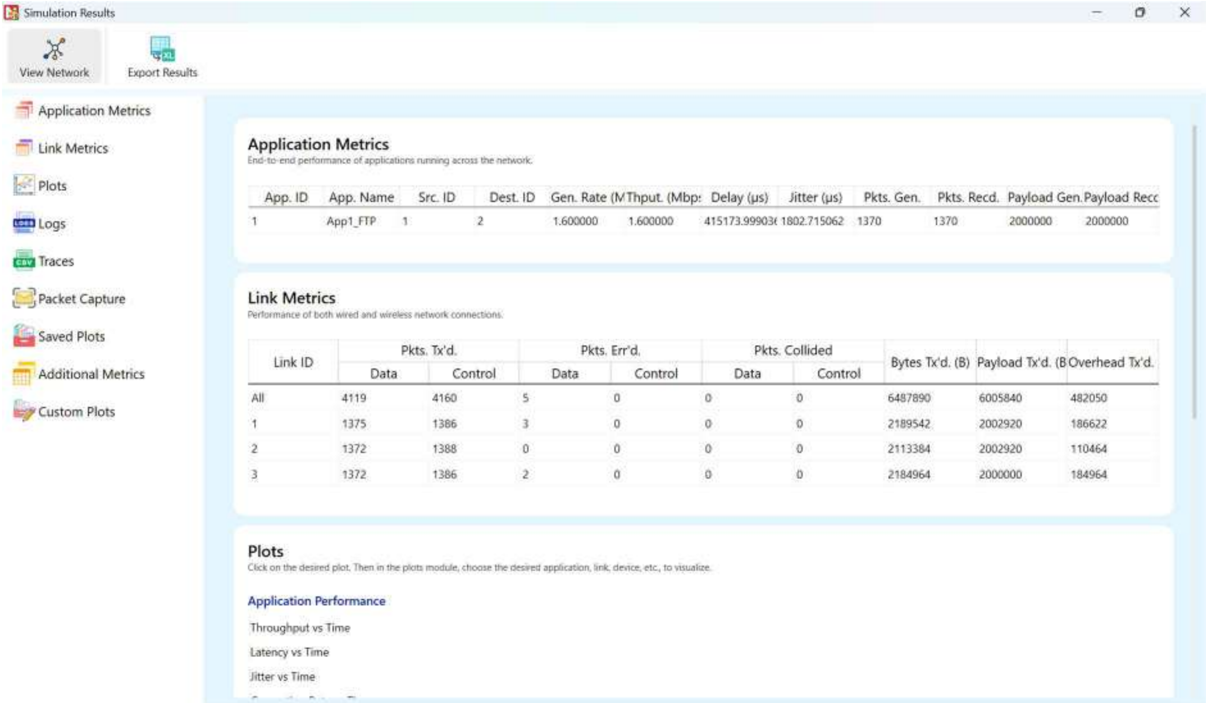
SETUP / PROCEDURE

1. Set up a UDP client and server in NetSim Lite.
2. Configure bandwidth, delay, and packet size.
3. Run tests under different network loads.
4. Record throughput, delay, and packet loss.

TOPOLOGY:



TESTCASE 1:



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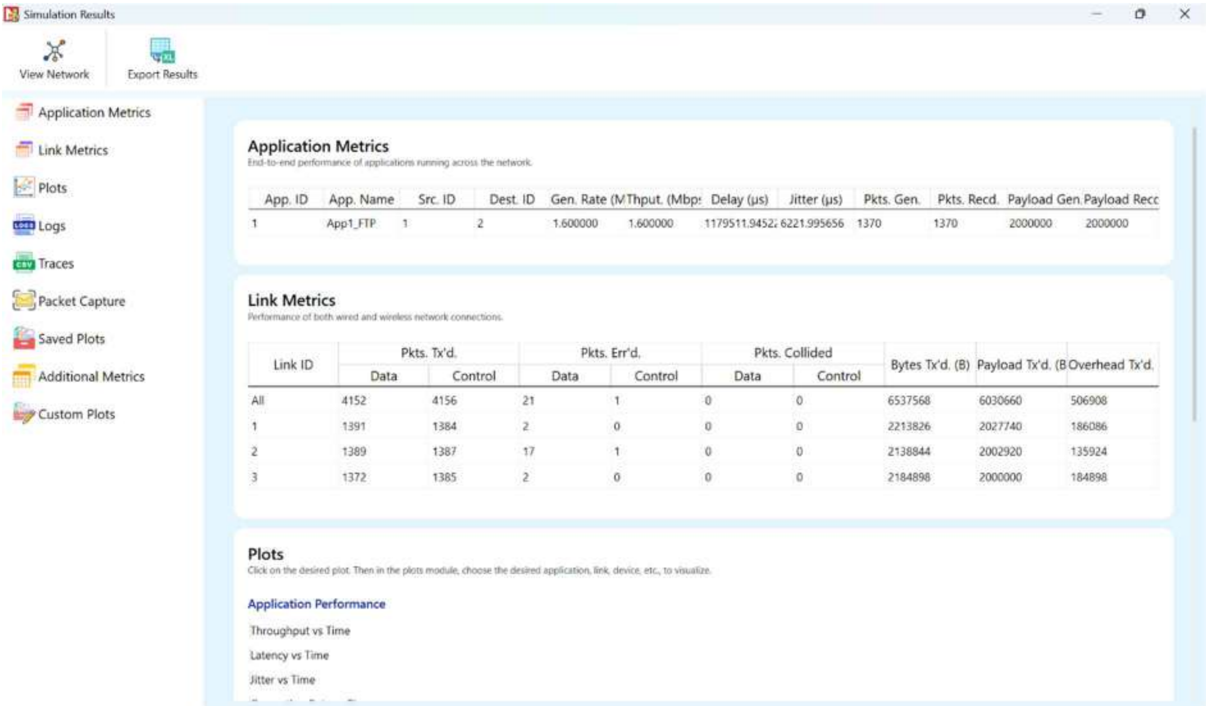
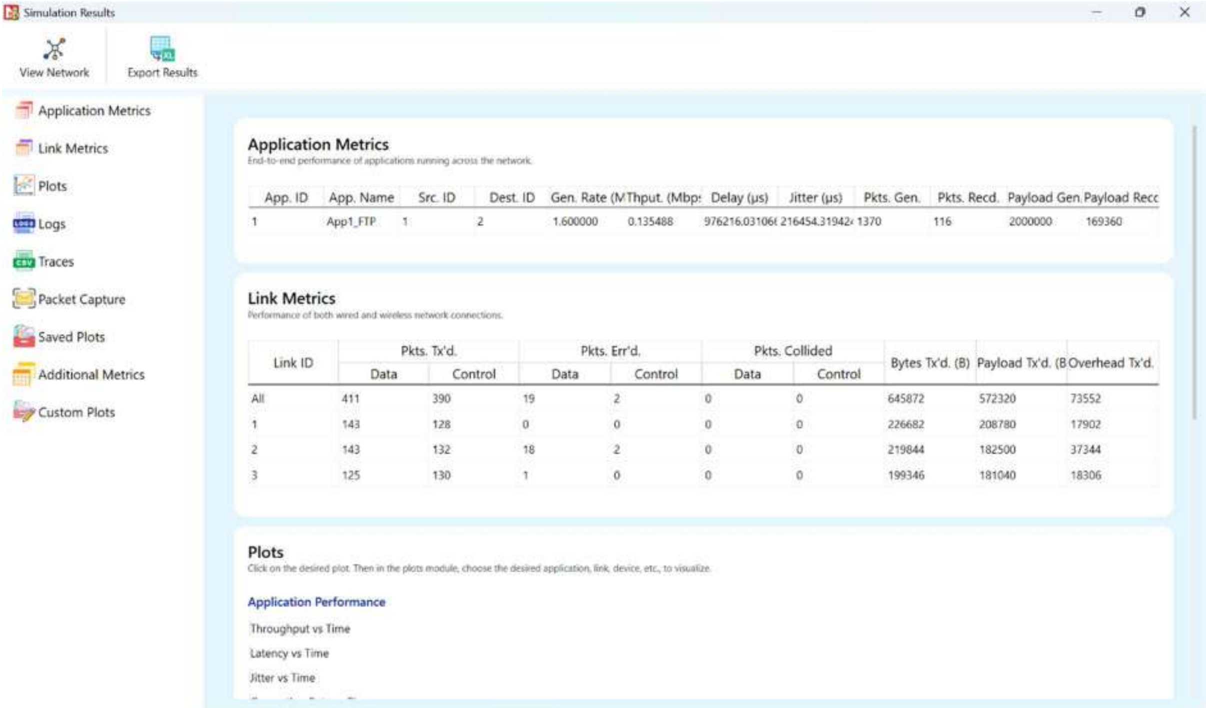


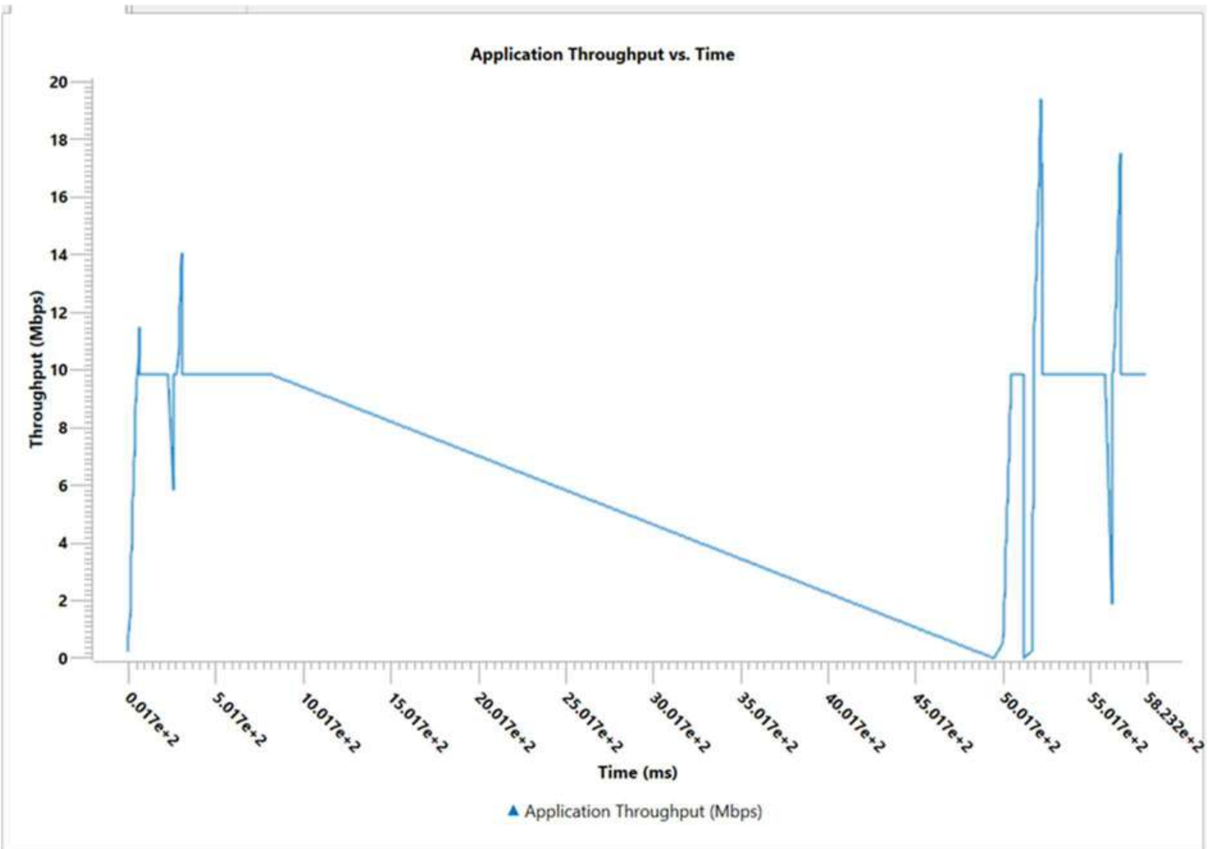
TESTCASE 3:

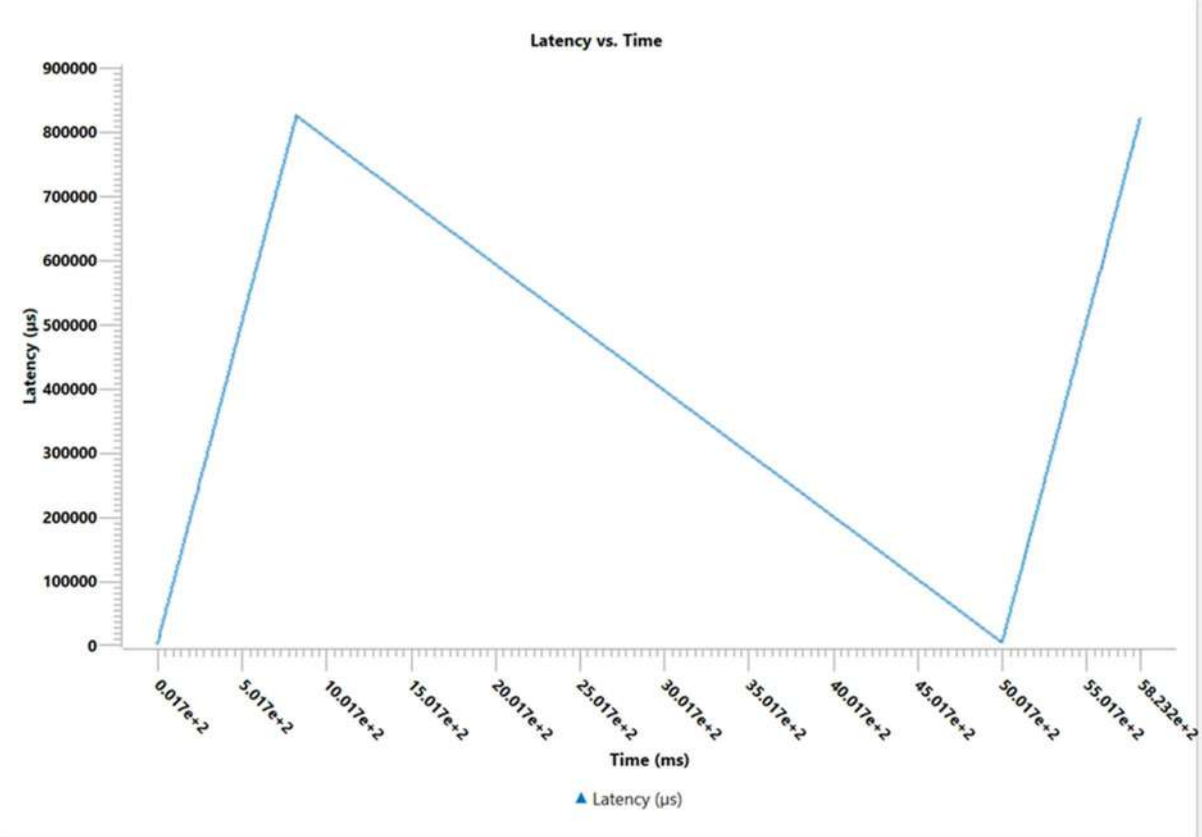


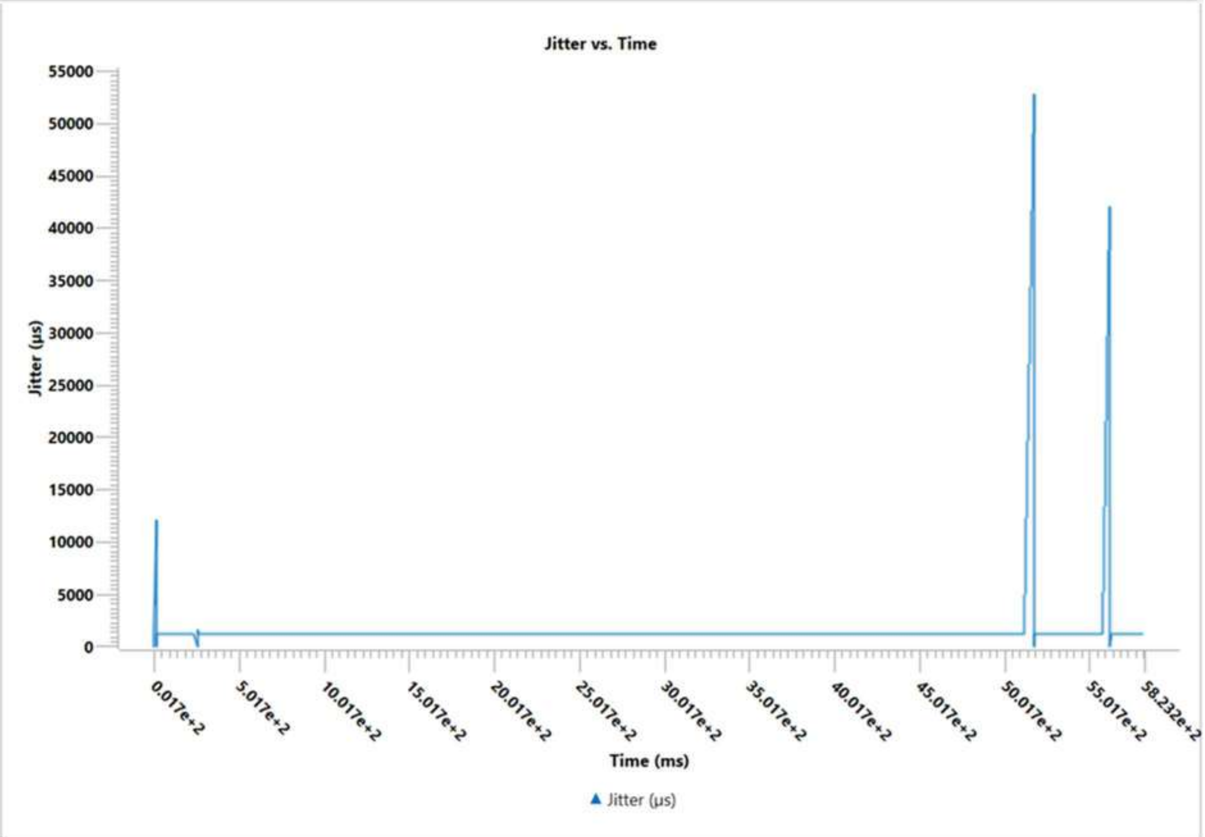
App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	1.600000	0.135488	976216.03106(	216454.31942‹	1370	116	2000000	169360

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	411	390	19	2	0	0	645872	572320	73552
1	143	128	0	0	0	0	226682	208780	17902
2	143	132	18	2	0	0	219844	182500	37344
3	125	130	1	0	0	0	199346	181040	18306

GRAPHS:







OBSERVATION

UDP achieves higher throughput than TCP but may lose packets in congested networks.

CONCLUSION

UDP offers faster transmission but sacrifices reliability, making it suitable for delay-sensitive data transfer.

Experiment 3: TCP under Packet Loss

Aim

To analyse TCP performance when packet loss occurs in the network.

Introduction

TCP uses retransmissions to recover from lost packets. While this maintains reliability, it reduces throughput and increases delay.

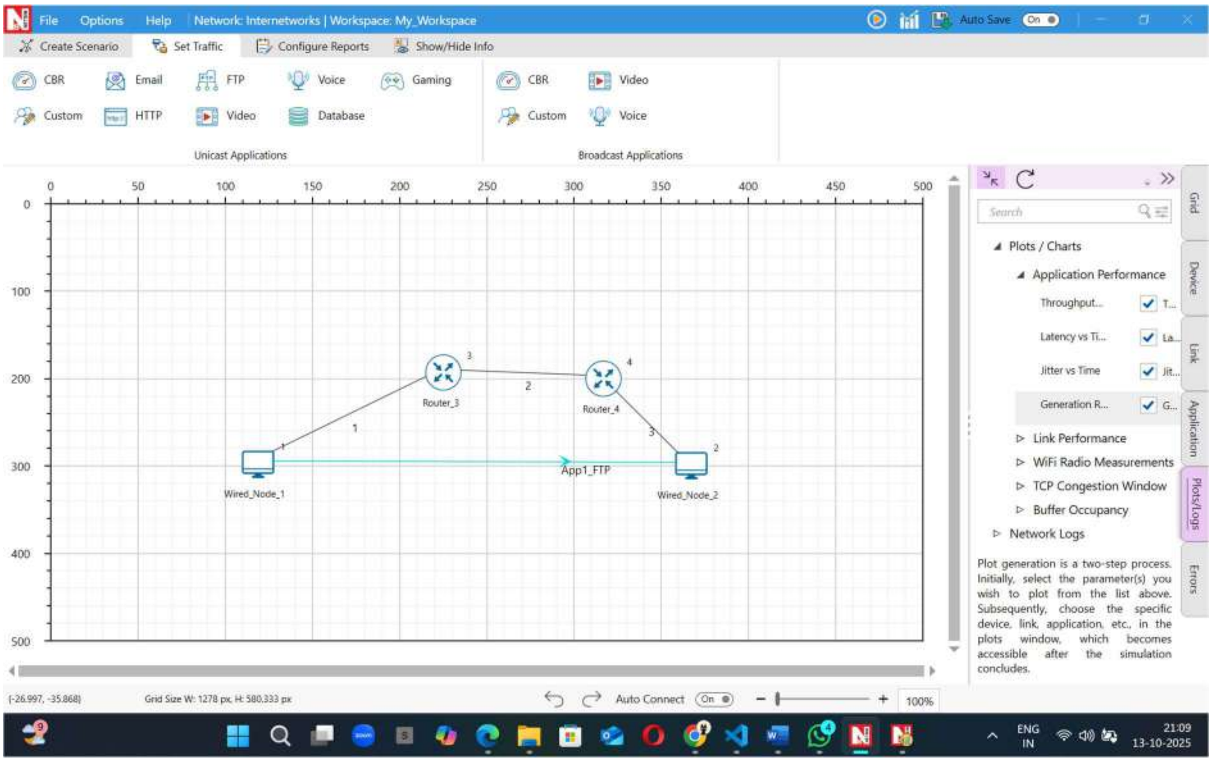
Theory

When packet loss occurs, TCP triggers retransmissions and enters congestion control phases. Increased packet loss causes TCP to slow its sending rate.

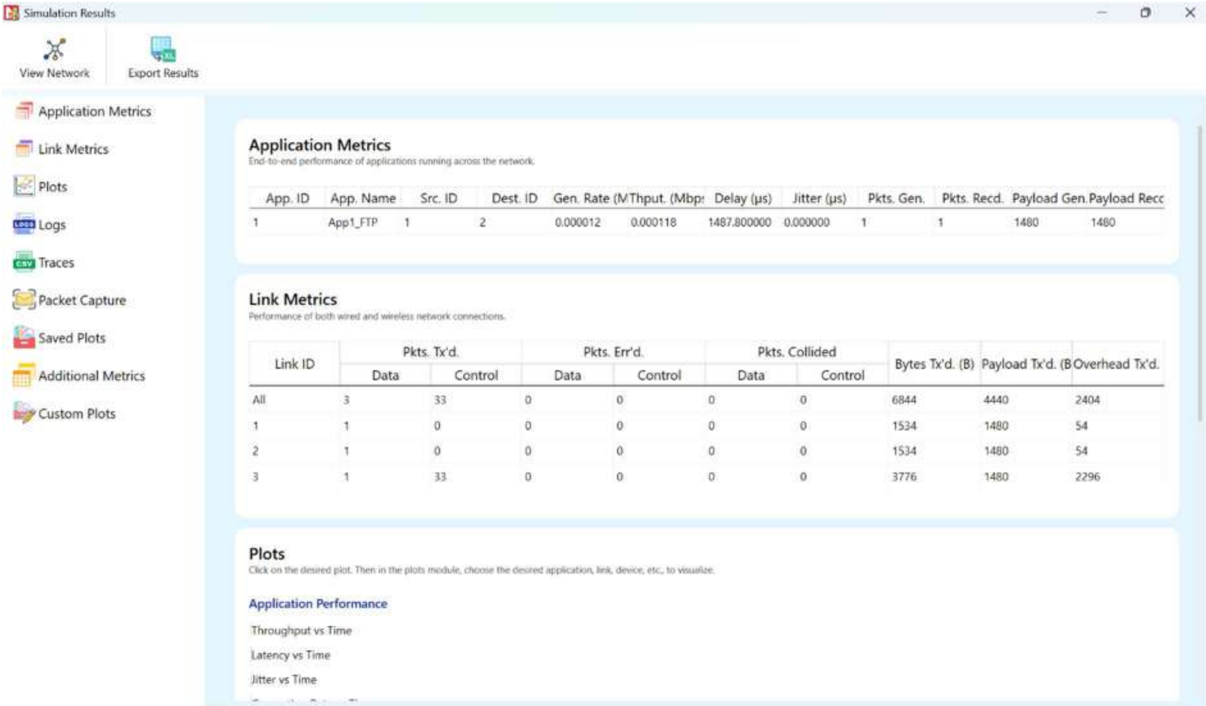
Setup / Procedure

1. Create a TCP network scenario with adjustable link error rate.
2. Set BER values (1%, 5%, 10%).
3. Observe retransmission counts, throughput, and transfer time.

TOPOLOGY:



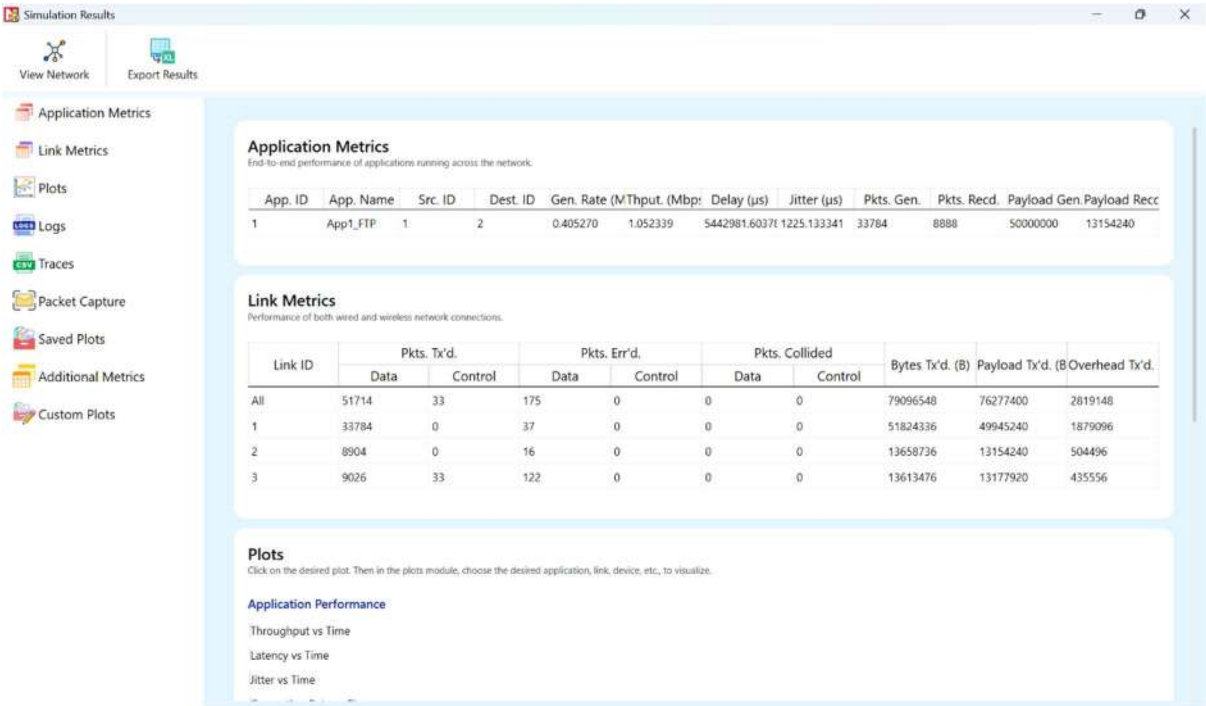
TESTCASE 1:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.000012	0.000118	1487.800000	0.000000	1	1	1480	1480

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	3	33	0	0	0	0	6844	4440	2404
1	1	0	0	0	0	0	1534	1480	54
2	1	0	0	0	0	0	1534	1480	54
3	1	33	0	0	0	0	3776	1480	2296

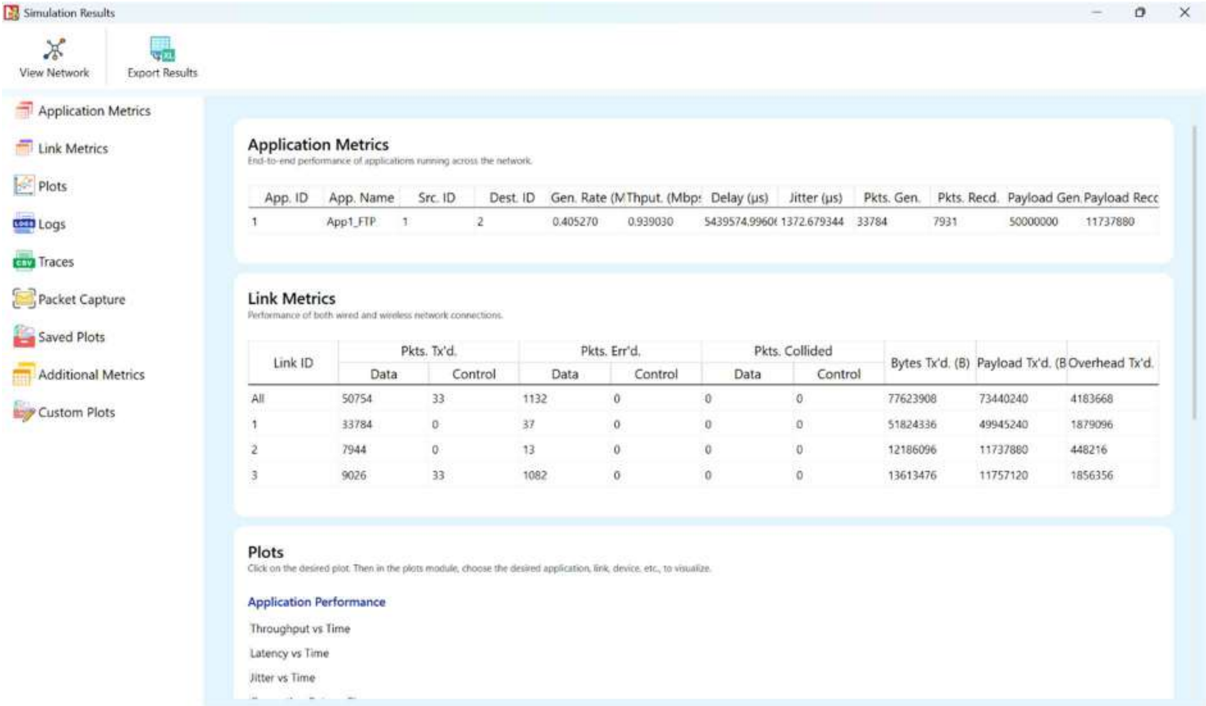
TESTCASE 2:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.405270	1.052339	5442981.6037(	1225.133341	33784	8888	50000000	13154240

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	51714	33	175	0	0	0	79096548	76277400	2819148
1	33784	0	37	0	0	0	51824336	49945240	1879096
2	8904	0	16	0	0	0	13658736	13154240	504496
3	9026	33	122	0	0	0	13613476	13177920	435556

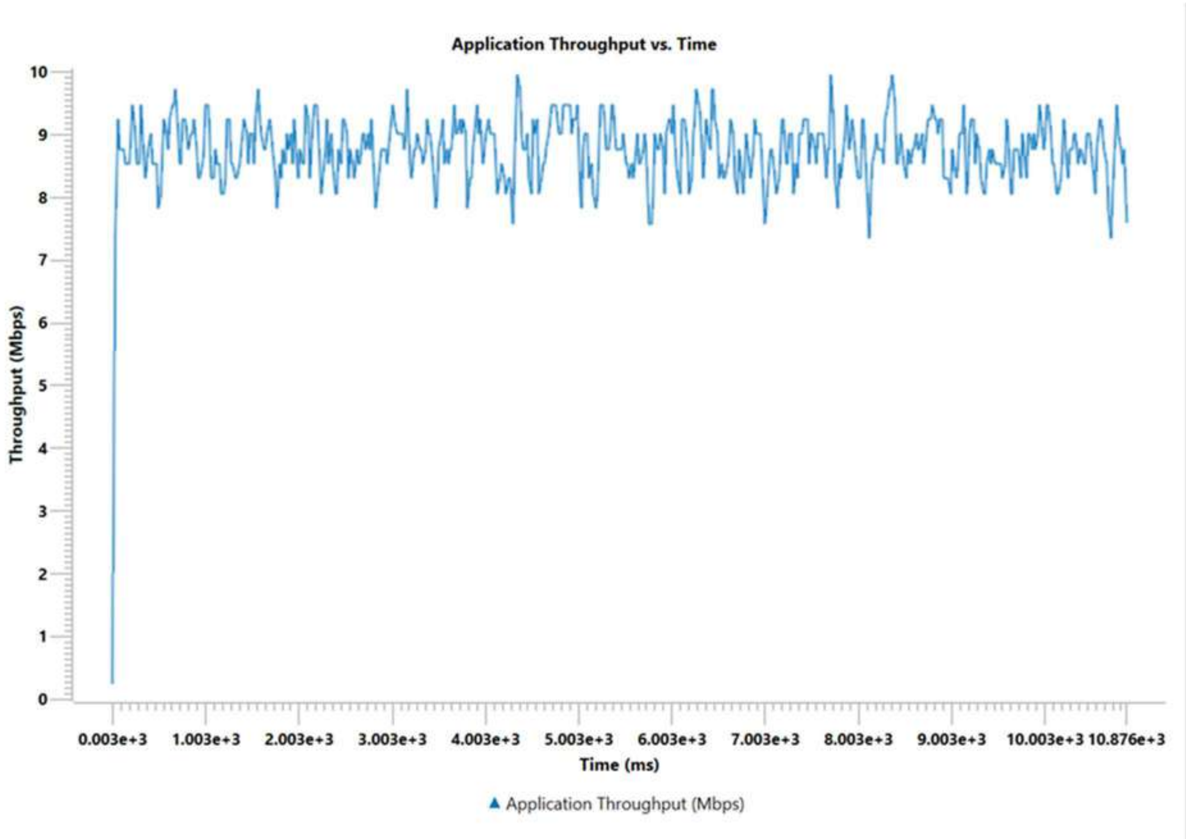
TESTCASE 3:

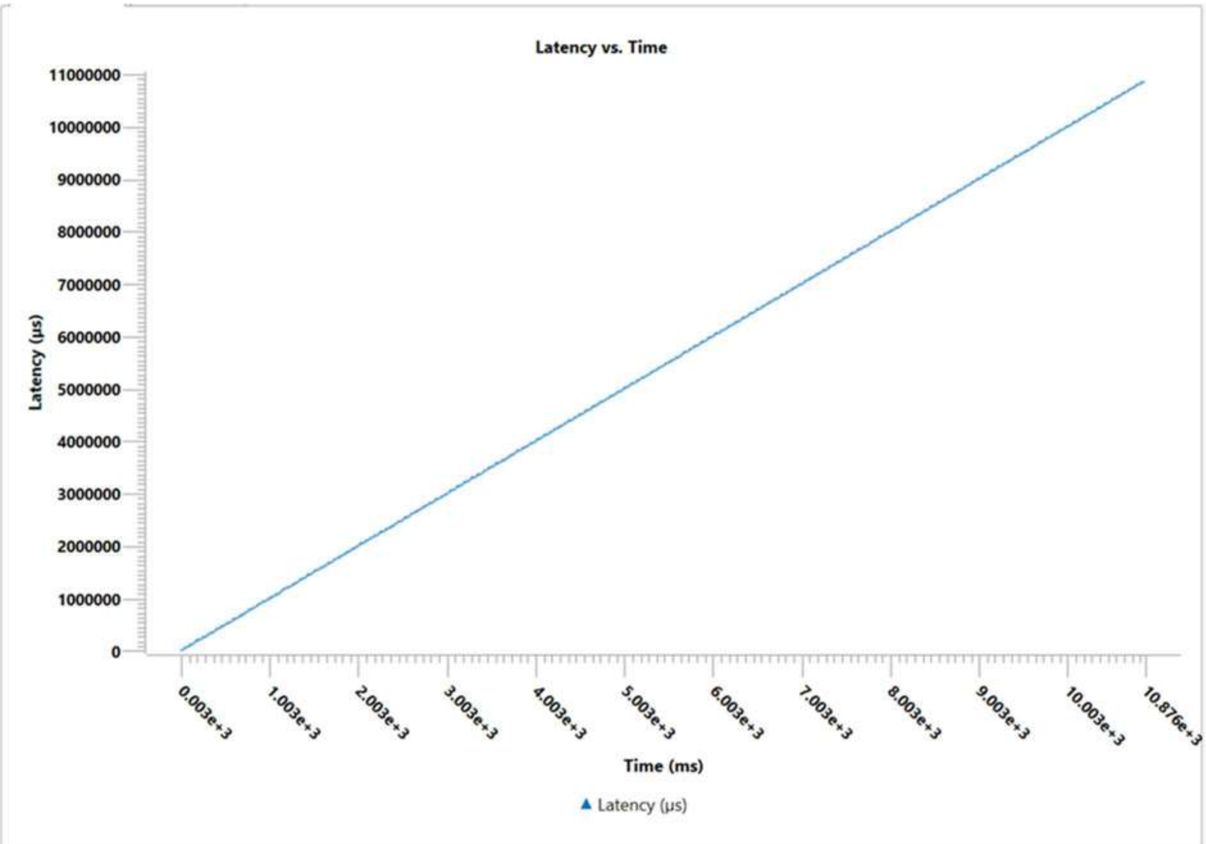


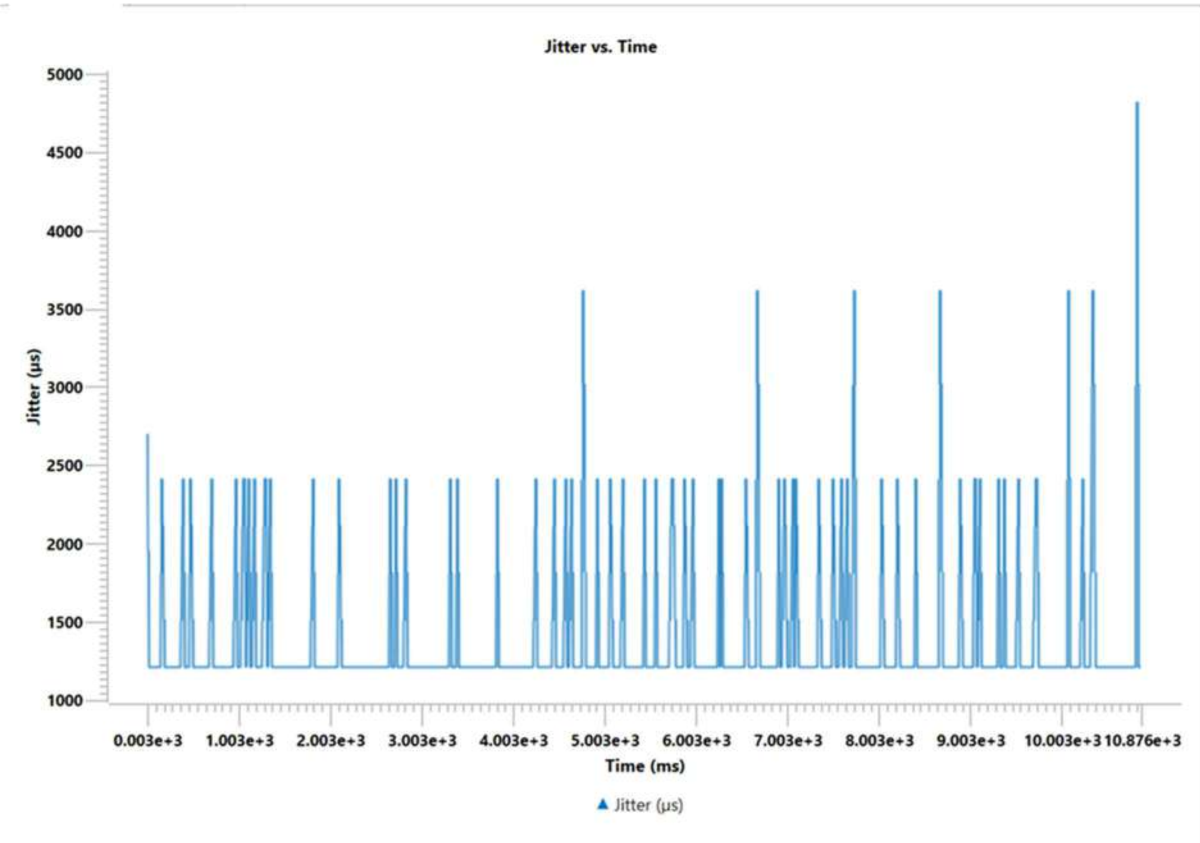
App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.405270	0.939030	5439574.9960(	1372.679344	33784	7931	50000000	11737880

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	50754	33	1132	0	0	0	77623908	73440240	4183668
1	33784	0	37	0	0	0	51824336	49945240	1879096
2	7944	0	13	0	0	0	12186096	11737880	448216
3	9026	33	1082	0	0	0	13613476	11757120	1856356

GRAPHS:




Latency vs. Time
Latency (µs)
11000000
10000000
9000000
8000000
7000000
6000000
5000000
4000000
3000000
2000000
1000000
0
0.003e+3
1.003e+3
2.003e+3
3.003e+3
4.003e+3
5.003e+3
6.003e+3
7.003e+3
8.003e+3
9.003e+3
10.003e+3
10.876e+3
Time (ms)
Latency (µs)


Jitter vs. Time
Jitter (µs)
5000
4500
4000
3500
3000
2500
2000
1500
1000
0.003e+3
1.003e+3
2.003e+3
3.003e+3
4.003e+3
5.003e+3
6.003e+3
7.003e+3
8.003e+3
9.003e+3
10.003e+3
10.876e+3
Time (ms)
Jitter (µs)

OBSERVATION

As packet loss increases, retransmissions rise, reducing throughput.

CONCLUSION

TCP maintains reliability at the cost of performance when packet loss occurs.

Experiment 4: UDP under Packet Loss

AIM

To study UDP performance in networks experiencing packet loss.

INTRODUCTION

Since UDP lacks retransmission and acknowledgment, packet loss directly reduces data accuracy and quality.

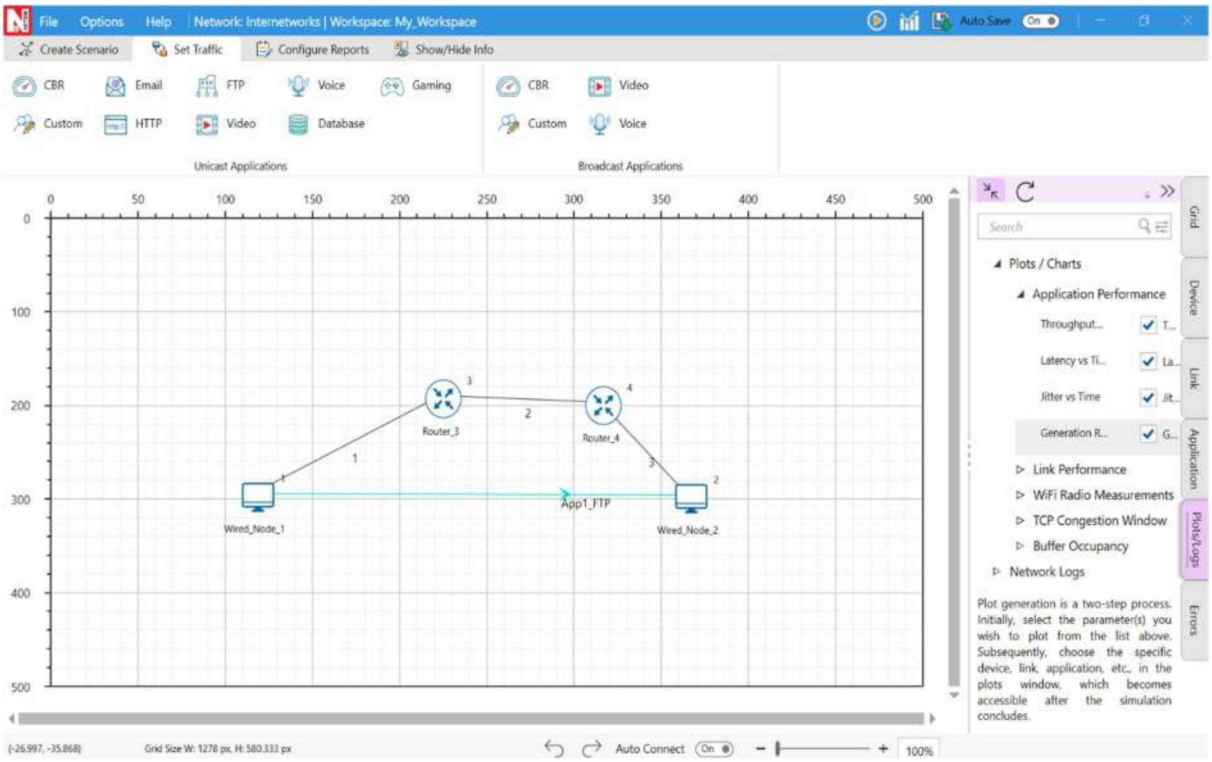
THEORY

UDP sends packets without error recovery. Any lost packet is permanently dropped, which affects real-time applications such as video or voice streaming.

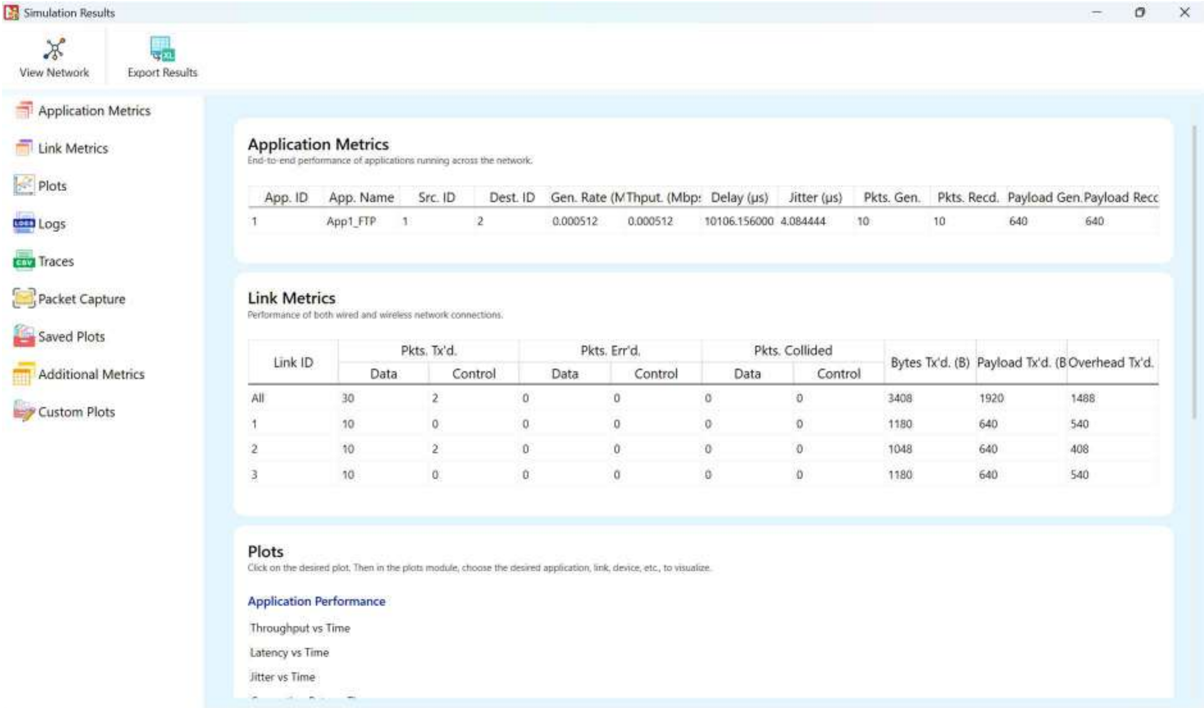
SETUP / PROCEDURE

1. Set up a UDP network and introduce different BER levels.
2. Record packet loss rate, delay, and jitter.
3. Compare results under increasing loss.

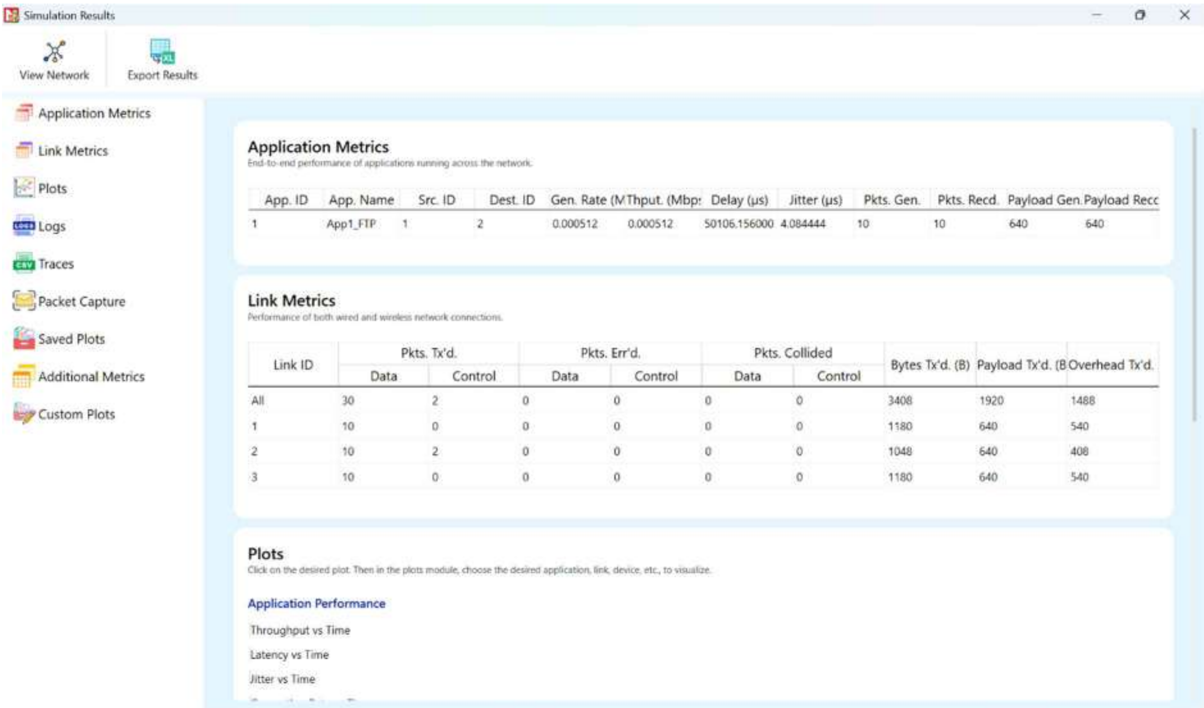
TOPOLOGY:



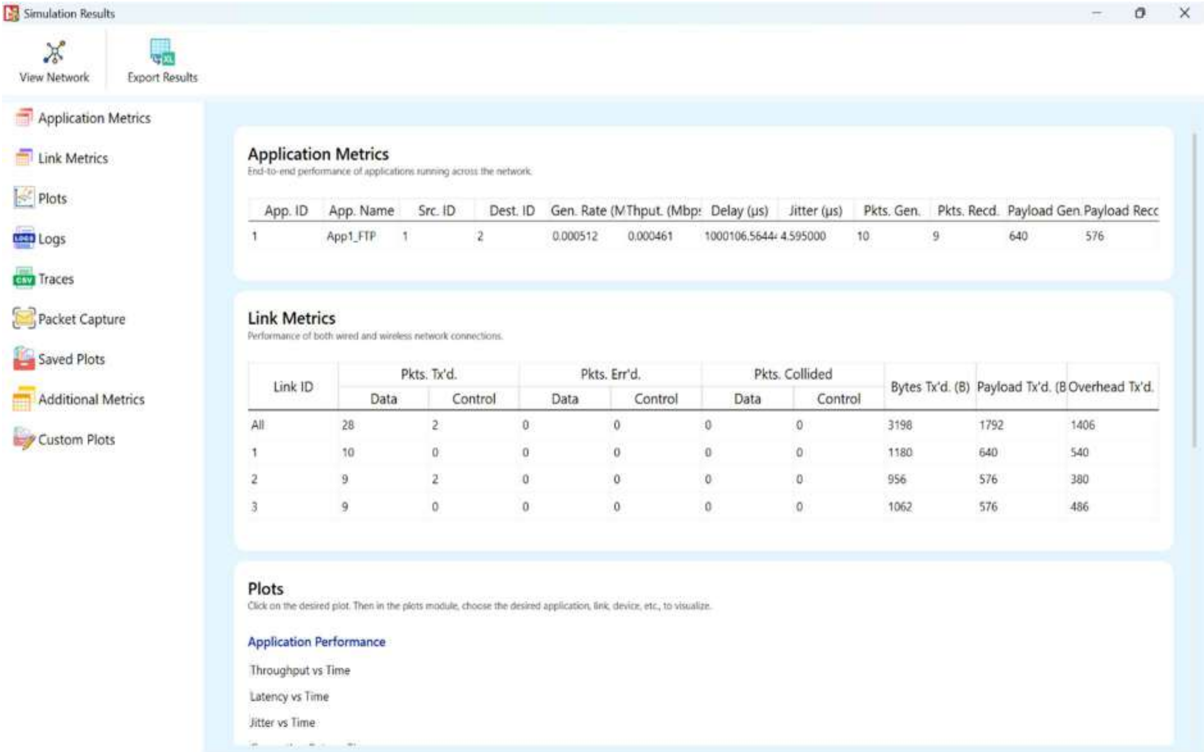
TESTCASE 1:



TESTCASE 2:



TESTCASE 3:


Simulation Results

View Network

Export Results

Application Metrics

Link Metrics

Plots

Logs

Traces

Packet Capture

Saved Plots

Additional Metrics

Custom Plots

Application Metrics

End-to-end performance of applications running across the network.

App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	2	0.000512	0.000461	1000106.5644	4.595000	10	9	640	576

Link Metrics

Performance of both wired and wireless network connections.

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	28	2	0	0	0	0	3198	1792	1406
1	10	0	0	0	0	0	1180	640	540
2	9	2	0	0	0	0	956	576	380
3	9	0	0	0	0	0	1062	576	486

Plots

Click on the desired plot. Then in the plots module, choose the desired application, link, device, etc., to visualize.

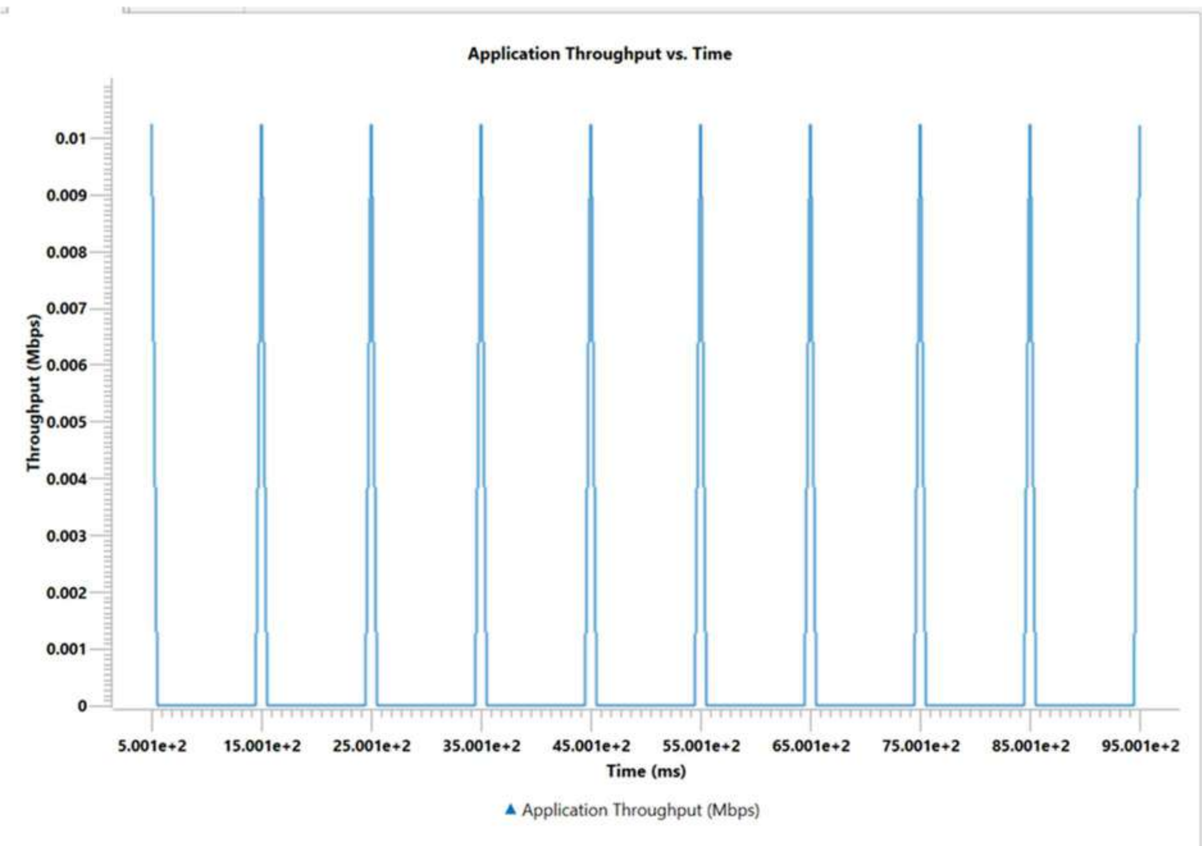
Application Performance

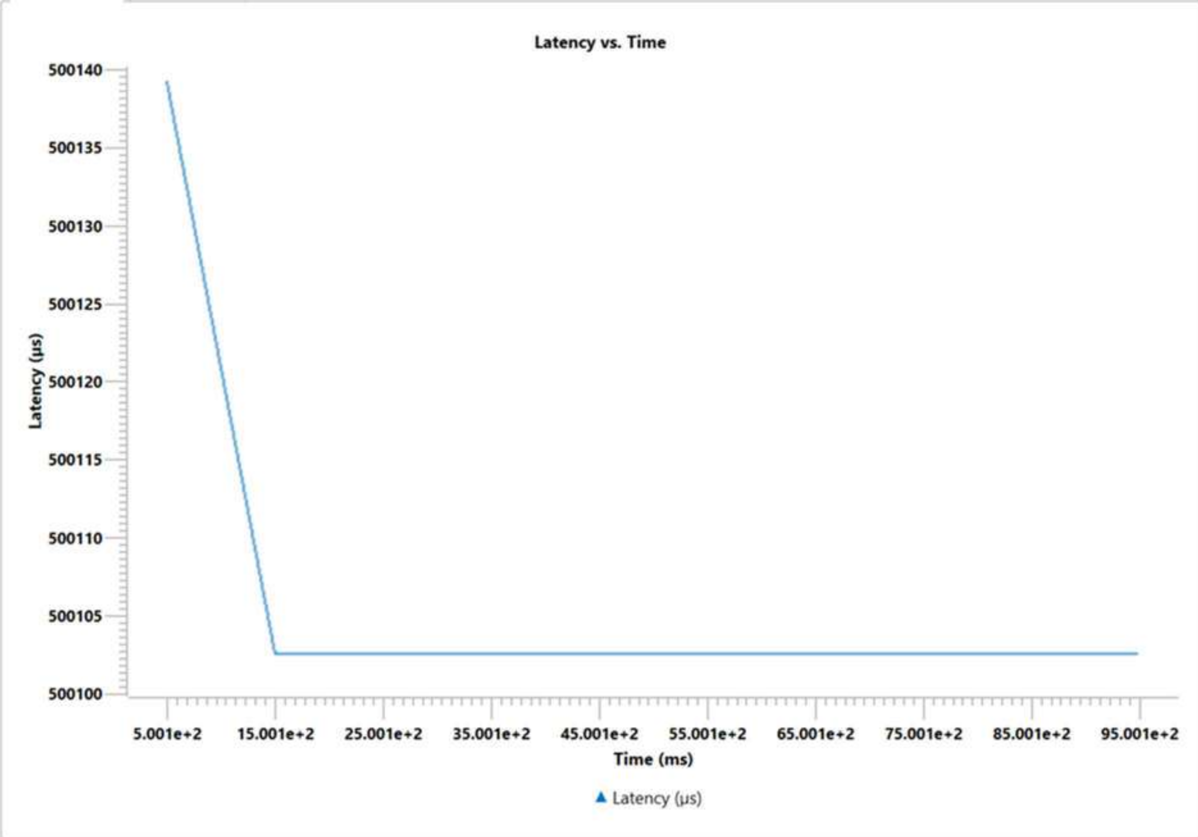
Throughput vs Time

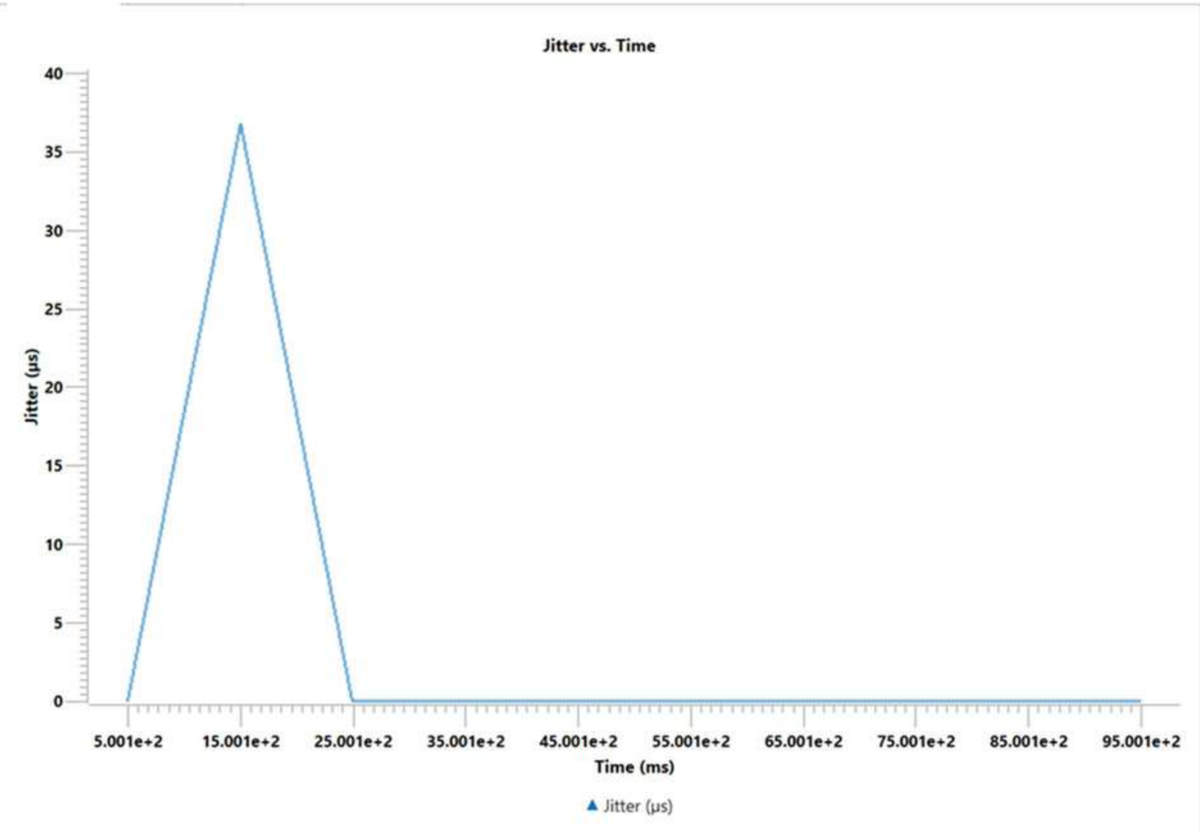
Latency vs Time

Jitter vs Time

GRAPHS:







OBSERVATION

Packet loss increases linearly with BER, with no retransmission recovery.

CONCLUSION

UDP performance degrades under packet loss, showing why it is best for tolerant real-time traffic.

Experiment 5: FTP Transfer Performance

AIM

To evaluate the performance of FTP (File Transfer Protocol) under various network conditions.

INTRODUCTION

FTP operates on top of TCP and provides reliable file transfer services. Its performance depends on the underlying TCP behaviour.

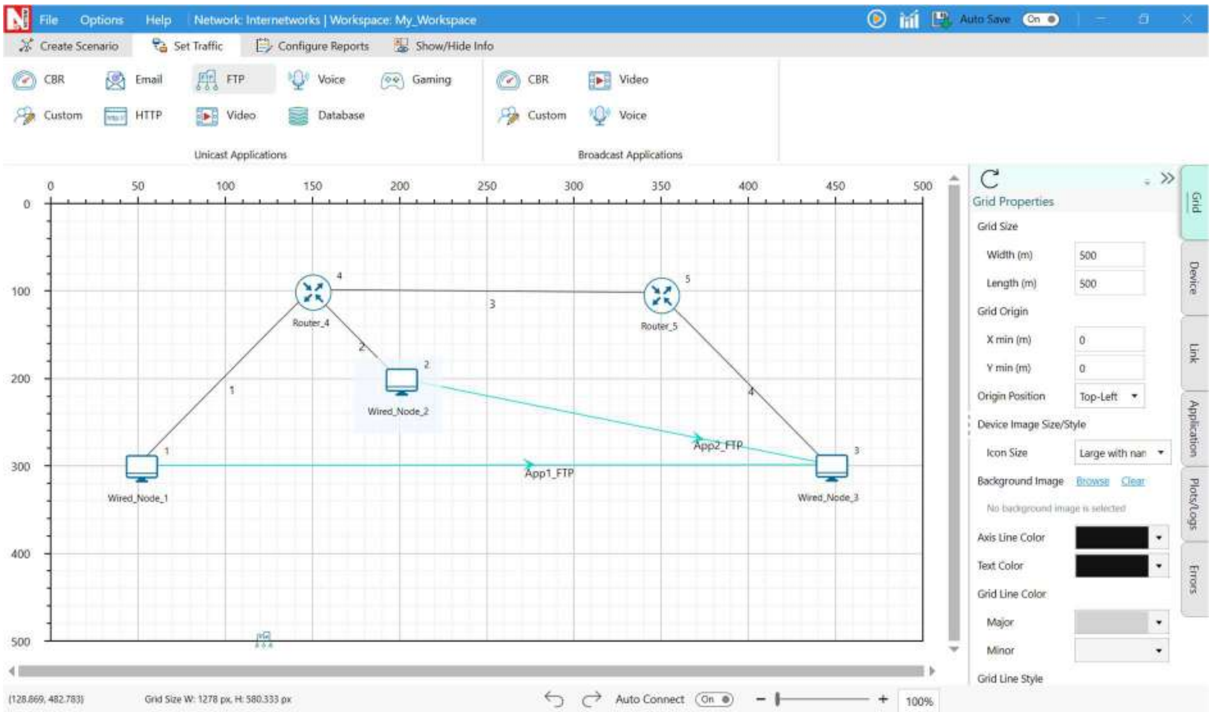
THEORY

Since FTP relies on TCP, factors like congestion, delay, and loss affect throughput. When congestion increases, FTP transfers become slower due to TCP's congestion avoidance mechanism.

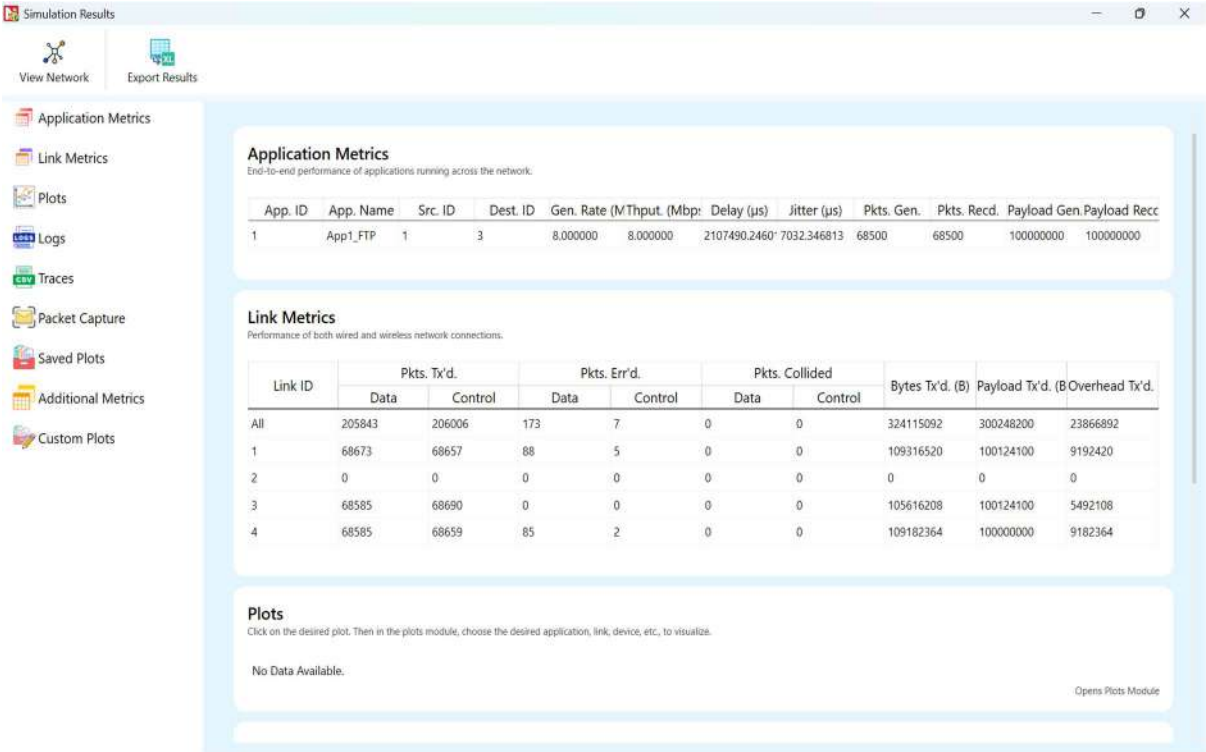
SETUP / PROCEDURE

1. Create an FTP client-server setup in NetSim Lite.
2. Vary bandwidth and background traffic.
3. Measure FTP throughput, delay, and transfer time.

TOPOLOGY



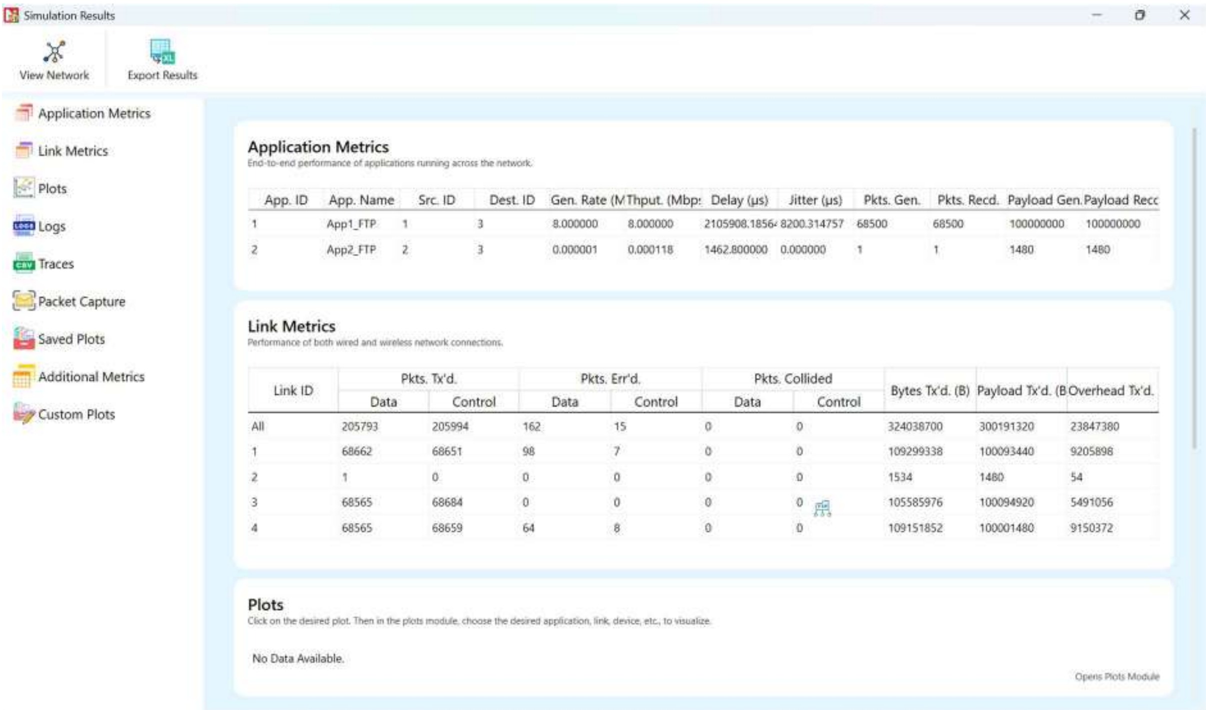
TESTCASE 1:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	3	8.000000	8.000000	2107490.2460	7032.346813	68500	68500	100000000	100000000

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	205843	206006	173	7	0	0	324115092	300248200	23866892
1	68673	68657	88	5	0	0	109316520	100124100	9192420
2	0	0	0	0	0	0	0	0	0
3	68585	68690	0	0	0	0	105616208	100124100	5492108
4	68585	68659	85	2	0	0	109182364	100000000	9182364

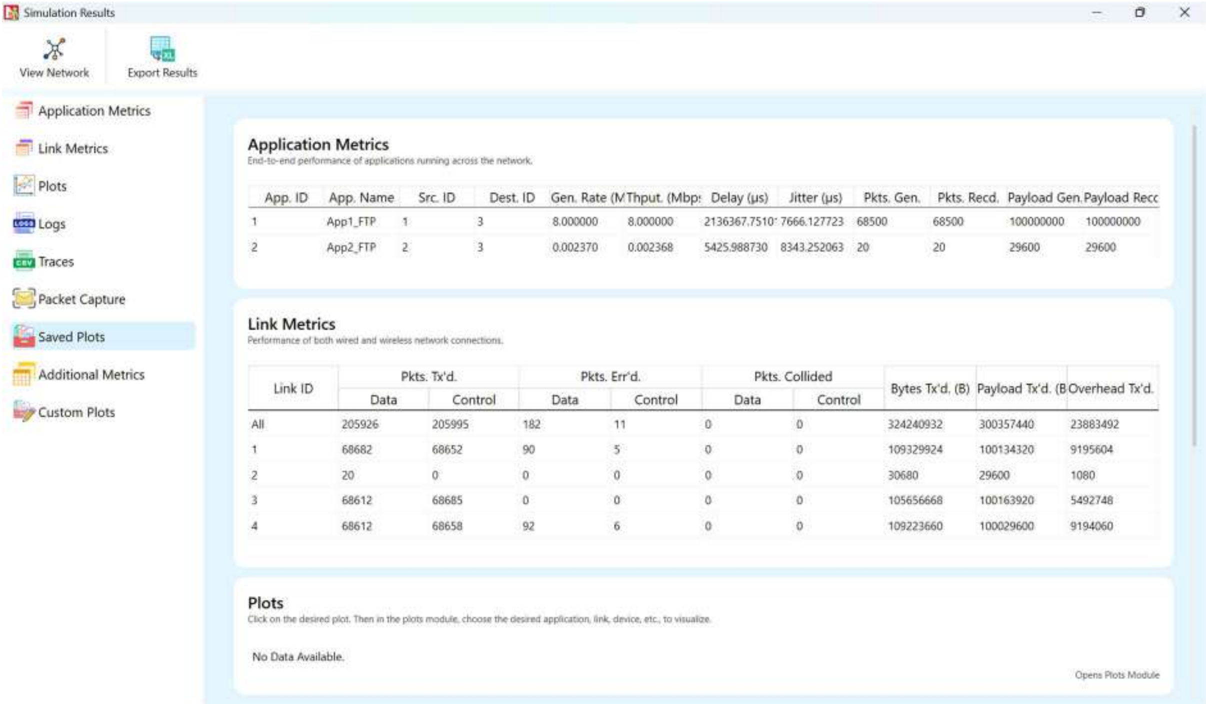
TESTCASE 2:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	3	8.000000	8.000000	2105908.1856	8200.314757	68500	68500	100000000	100000000
2	App2_FTP	2	3	0.000001	0.000118	1462.800000	0.000000	1	1	1480	1480

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	205793	205994	162	15	0	0	324038700	300191320	23847380
1	68662	68651	98	7	0	0	109299338	100093440	9205898
2	1	0	0	0	0	0	1534	1480	54
3	68565	68684	0	0	0	0	105585976	100094920	5491056
4	68565	68659	64	8	0	0	109151852	100001480	9150372

TESTCASE 3:



Application Metrics

End-to-end performance of applications running across the network.

App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	3	8.000000	8.000000	2136367.7510	7666.127723	68500	68500	100000000	100000000
2	App2_FTP	2	3	0.002370	0.002368	5425.988730	8343.252063	20	20	29600	29600

Link Metrics

Performance of both wired and wireless network connections.

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	205926	205995	182	11	0	0	324240932	300357440	23883492
1	68682	68652	90	5	0	0	109329924	100134320	9195604
2	20	0	0	0	0	0	30680	29600	1080
3	68612	68685	0	0	0	0	105656668	100163920	5492748
4	68612	68658	92	6	0	0	109223660	100029600	9194060

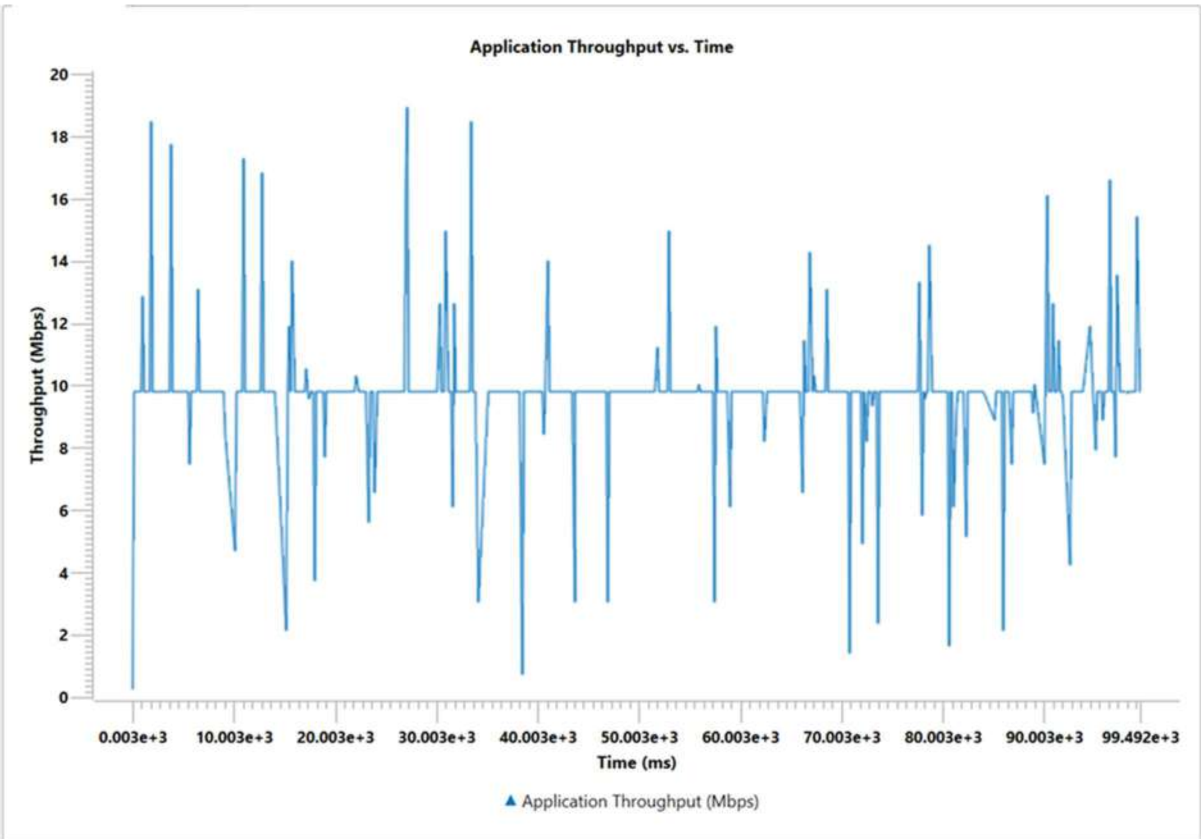
Plots

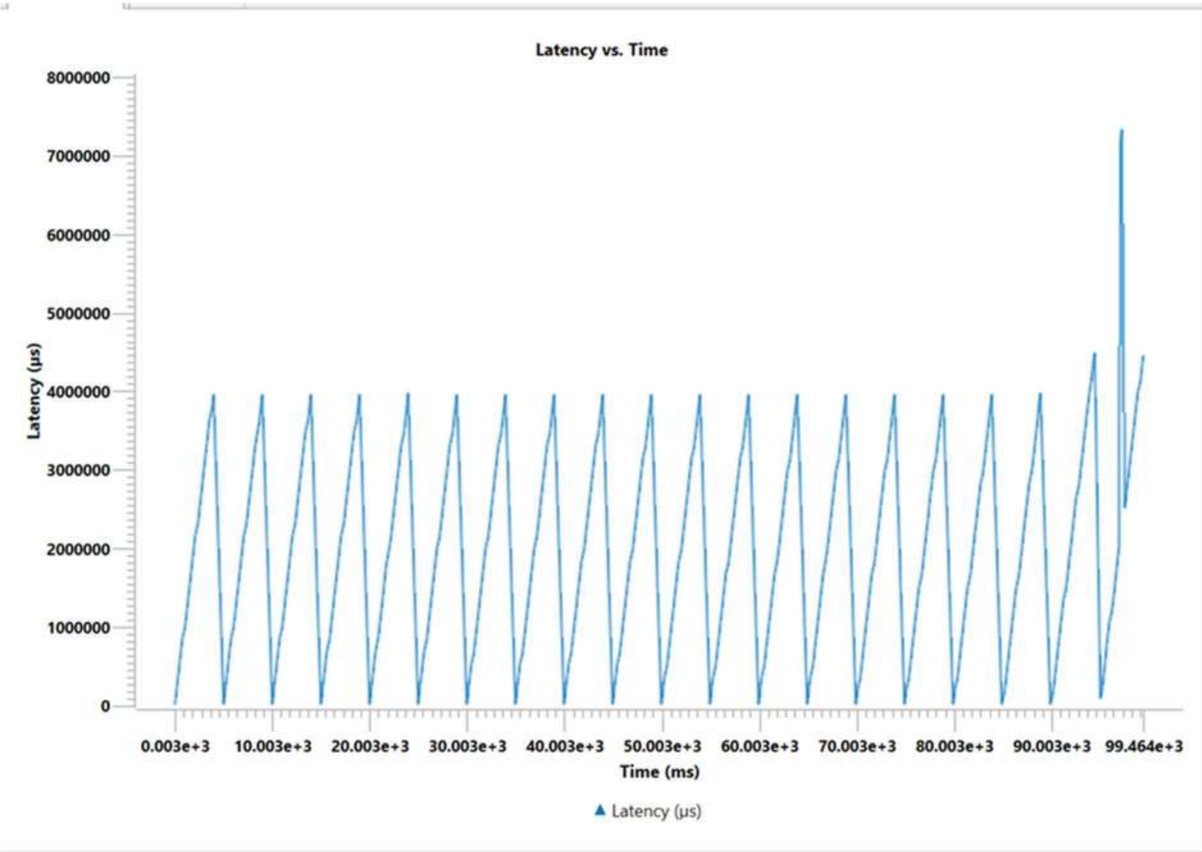
Click on the desired plot. Then in the plots module, choose the desired application, link, device, etc., to visualize.

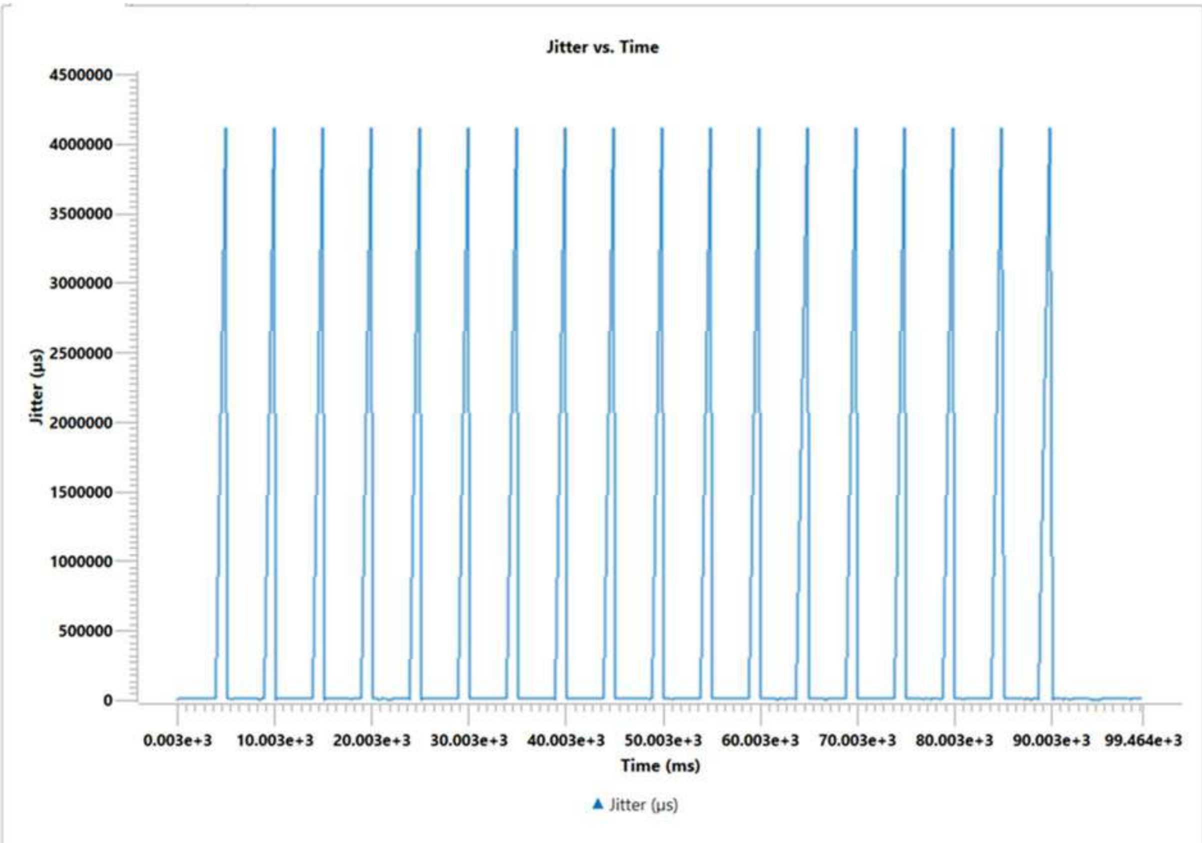
No Data Available.

Opens Plots Module

GRAPHS :







OBSERVATION

FTP throughput decreases during congestion or high delay.

CONCLUSION

FTP achieves reliable delivery, but performance declines when TCP experiences congestion.

Experiment 6: HTTP Web Browsing Performance

AIM

To analyse the performance of HTTP-based web browsing in different network conditions.

INTRODUCTION

HTTP runs on TCP and retrieves web resources such as HTML, images, and scripts. Its performance depends on network latency and number of parallel connections.

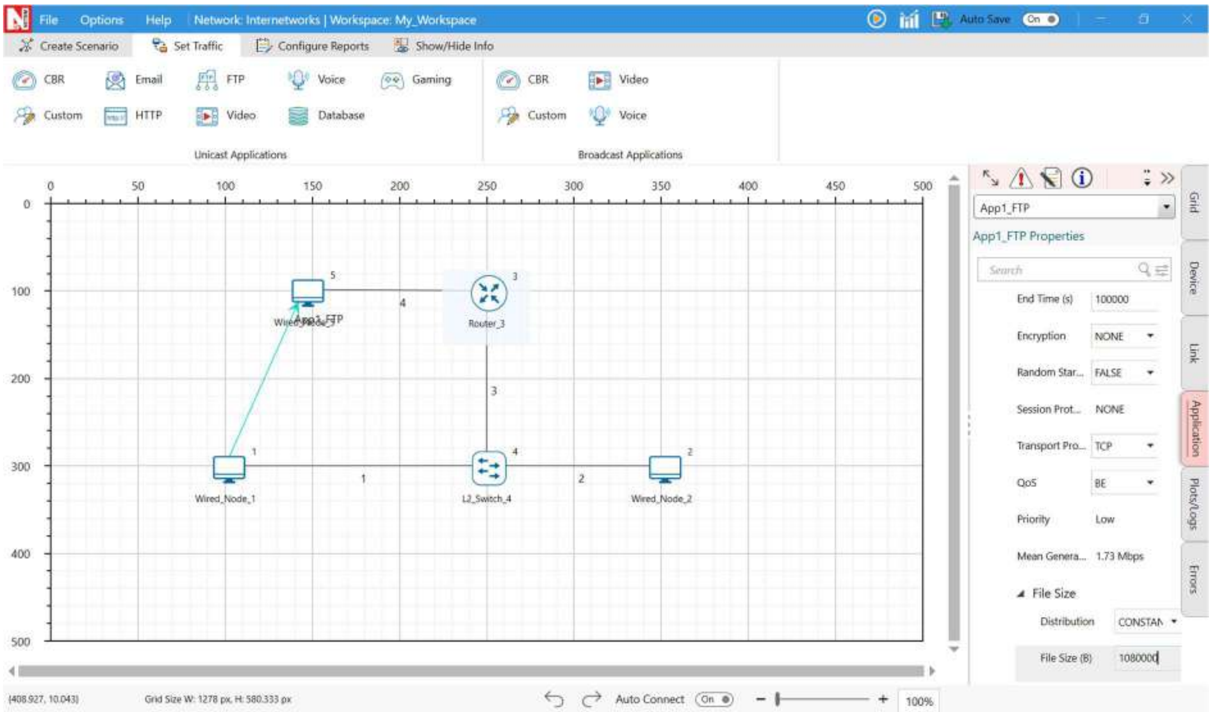
THEORY

Each HTTP request involves a TCP handshake and data transfer. Multiple connections (HTTP/1.1 or 2.0) can improve performance but increase overhead.

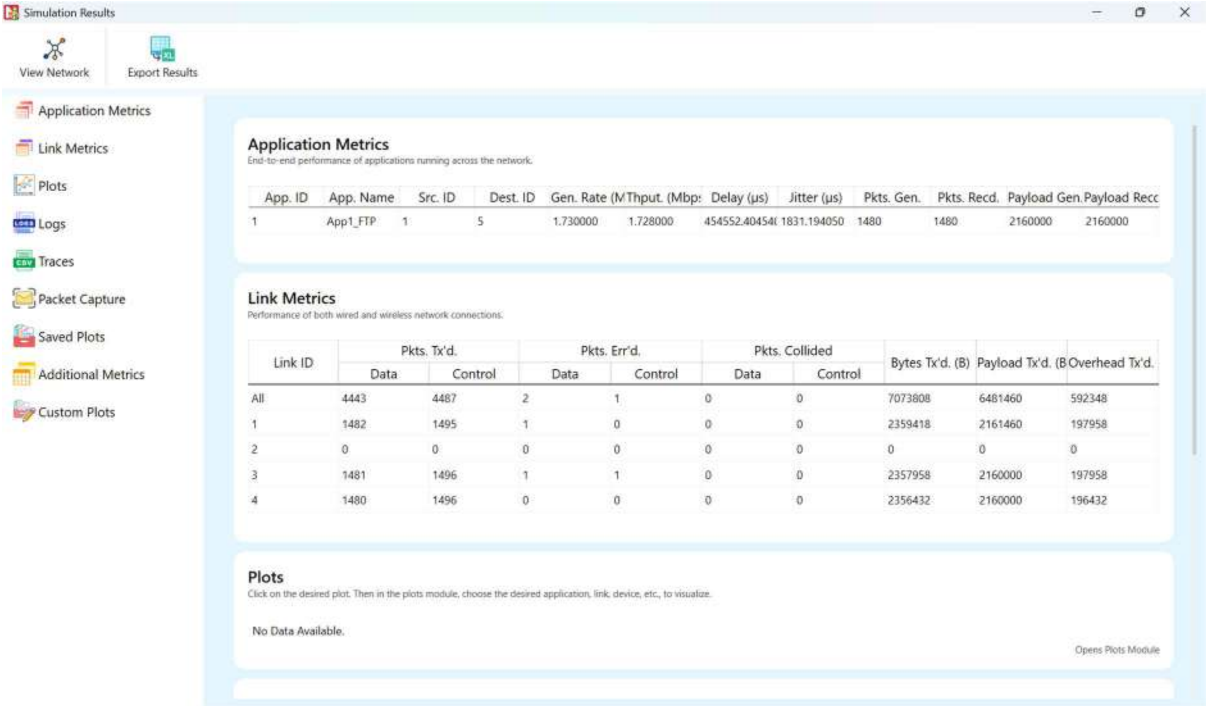
SETUP / PROCEDURE

1. Configure an HTTP client-server scenario in NetSim.
2. Simulate one large transfer vs. multiple small transfers.
3. Measure total transfer time and delay.

TOPOLOGY



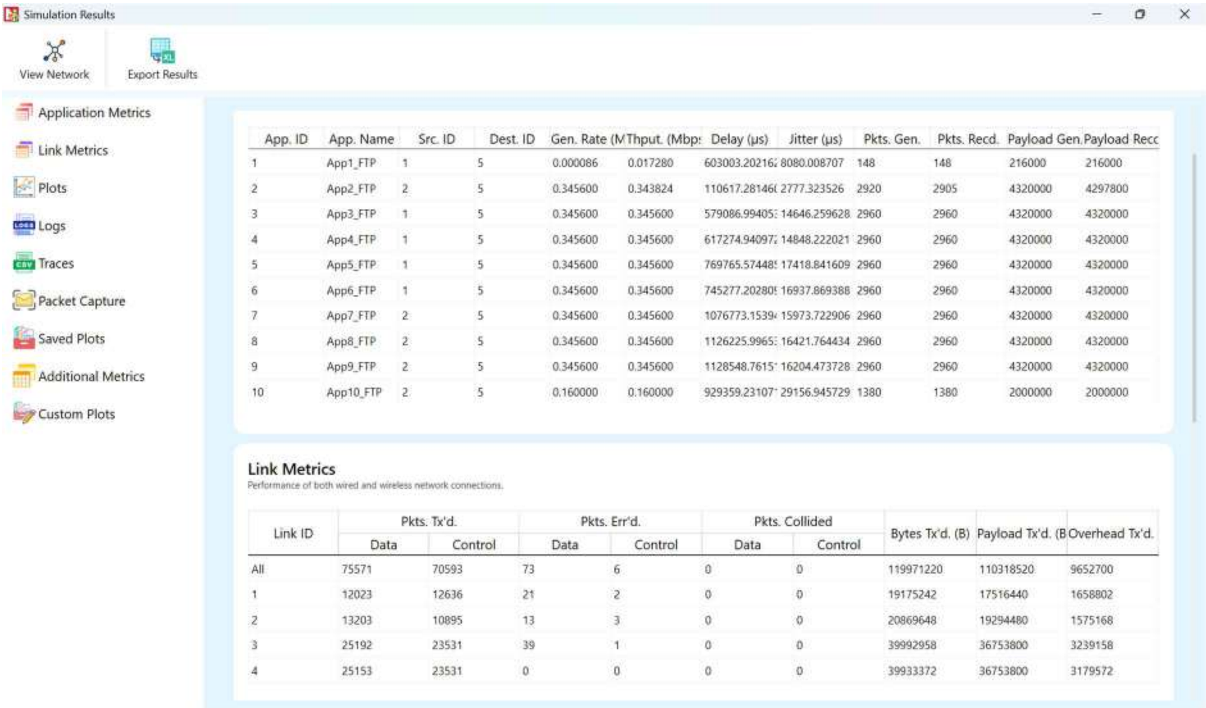
TESTCASE 1:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	5	1.730000	1.728000	454552.40454(	1831.194050	1480	1480	2160000	2160000

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	4443	4487	2	1	0	0	7073808	6481460	592348
1	1482	1495	1	0	0	0	2359418	2161460	197958
2	0	0	0	0	0	0	0	0	0
3	1481	1496	1	1	0	0	2357958	2160000	197958
4	1480	1496	0	0	0	0	2356432	2160000	196432

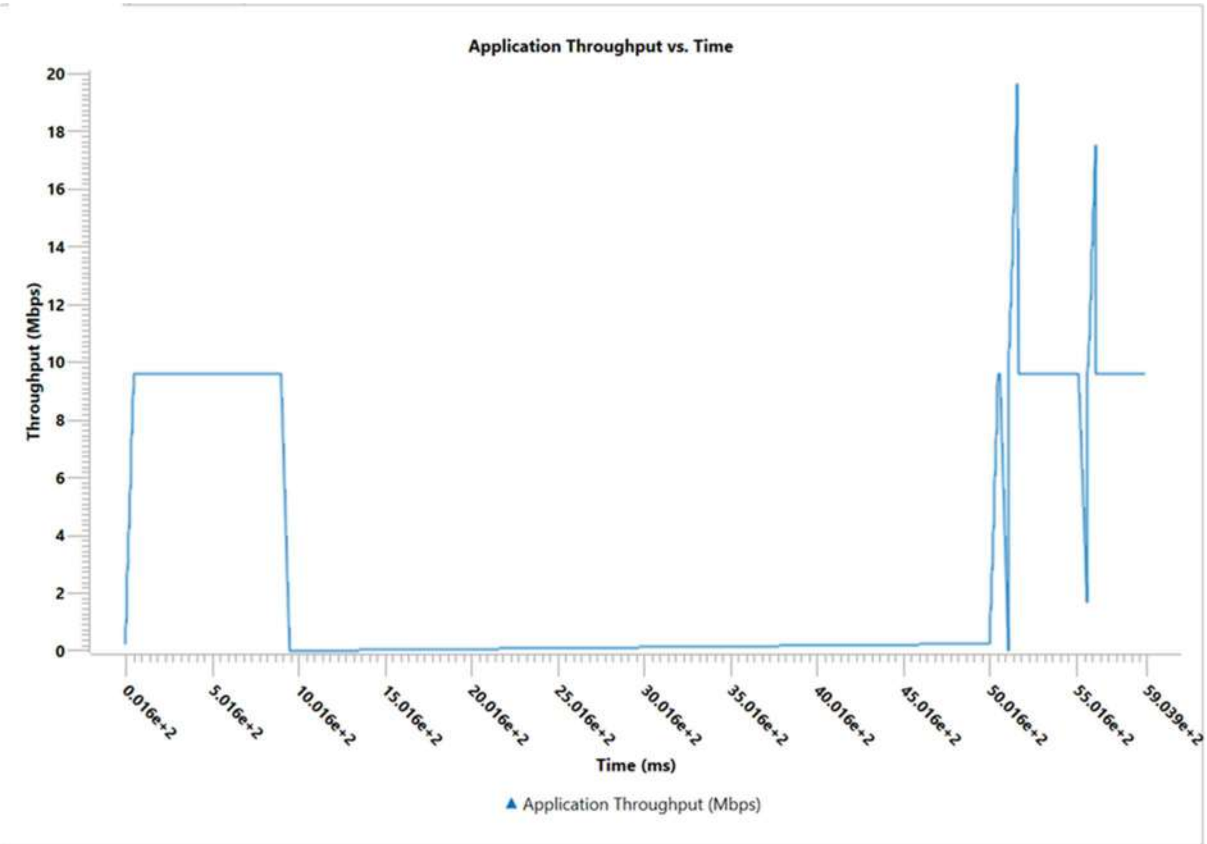
TESTCASE 2:

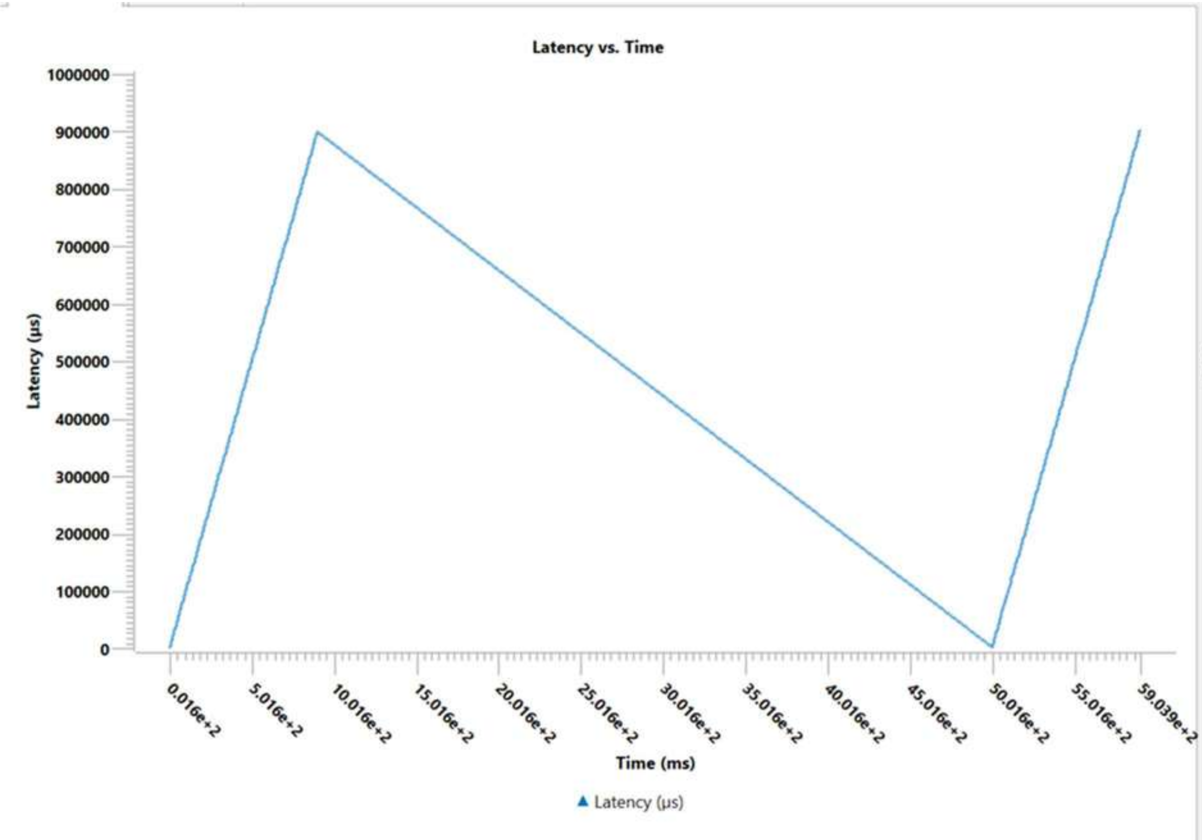


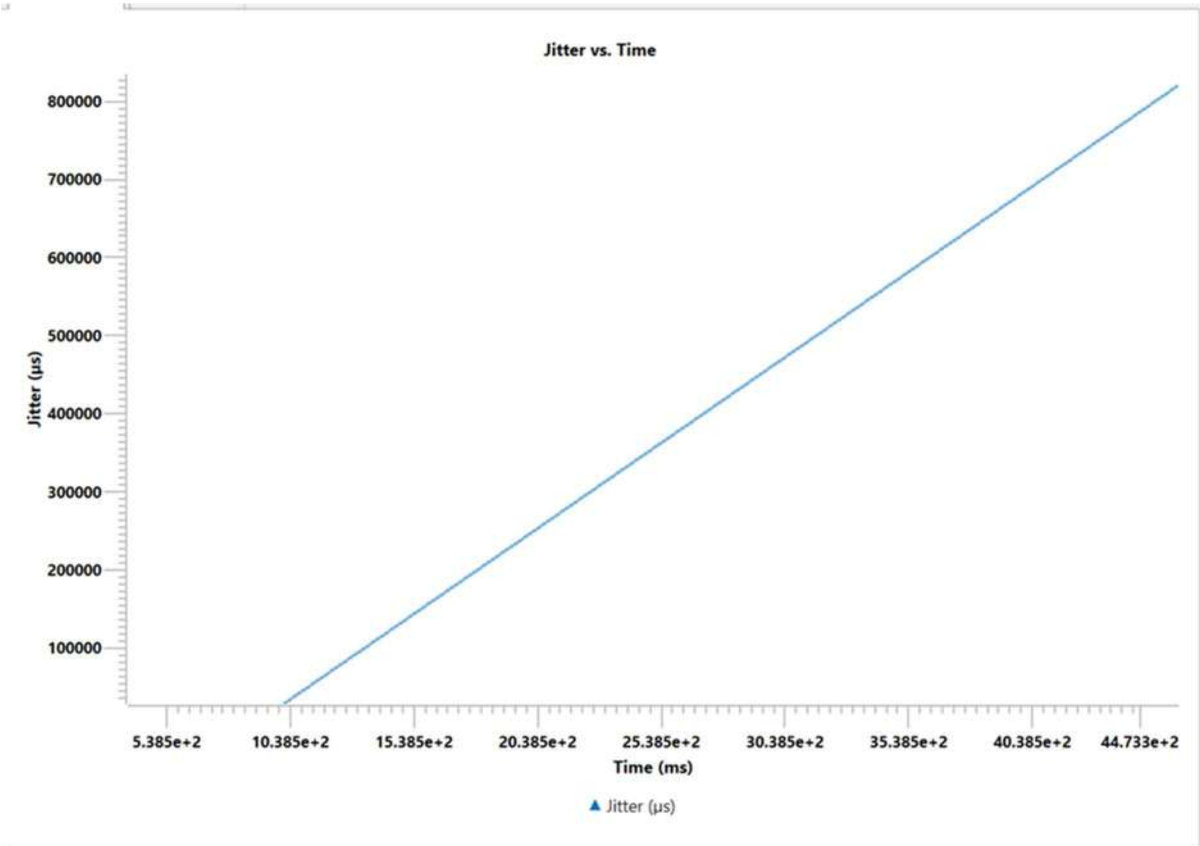
App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	5	0.000086	0.017280	603003.20216:	8080.008707	148	148	216000	216000
2	App2_FTP	2	5	0.345600	0.343824	110617.28146(	2777.323526	2920	2905	4320000	4297800
3	App3_FTP	1	5	0.345600	0.345600	579086.99405:	14646.259628	2960	2960	4320000	4320000
4	App4_FTP	1	5	0.345600	0.345600	617274.94097:	14848.222021	2960	2960	4320000	4320000
5	App5_FTP	1	5	0.345600	0.345600	769765.57448!	17418.841609	2960	2960	4320000	4320000
6	App6_FTP	1	5	0.345600	0.345600	745277.20280!	16937.869388	2960	2960	4320000	4320000
7	App7_FTP	2	5	0.345600	0.345600	1076773.1539	15973.722906	2960	2960	4320000	4320000
8	App8_FTP	2	5	0.345600	0.345600	1126225.9965:	16421.764434	2960	2960	4320000	4320000
9	App9_FTP	2	5	0.345600	0.345600	1128548.7615'	16204.473728	2960	2960	4320000	4320000
10	App10_FTP	2	5	0.160000	0.160000	929359.23107'	29156.945729	1380	1380	2000000	2000000

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	75571	70593	73	6	0	0	119971220	110318520	9652700
1	12023	12636	21	2	0	0	19175242	17516440	1658802
2	13203	10895	13	3	0	0	20869648	19294480	1575168
3	25192	23531	39	1	0	0	39992958	36753800	3239158
4	25153	23531	0	0	0	0	39933372	36753800	3179572

GRAPHS:


Application Throughput vs. Time
Throughput (Mbps)
20
18
16
14
12
10
8
6
4
2
0
0.016e+2
5.016e+2
10.016e+2
15.016e+2
20.016e+2
25.016e+2
30.016e+2
35.016e+2
40.016e+2
45.016e+2
50.016e+2
55.016e+2
59.039e+2
Time (ms)
Application Throughput (Mbps)


Latency vs. Time
Latency (µs)
1000000
900000
800000
700000
600000
500000
400000
300000
200000
100000
0
0.016e+2
5.016e+2
10.016e+2
15.016e+2
20.016e+2
25.016e+2
30.016e+2
35.016e+2
40.016e+2
45.016e+2
50.016e+2
55.016e+2
59.039e+2
Time (ms)
Latency (µs)



OBSERVATION

Single large transfers are efficient, while multiple small connections cause extra setup delay.

CONCLUSION

HTTP performance improves with fewer connections and lower latency.

Experiment 7: VoIP over TCP vs UDP

AIM

To compare VoIP performance using TCP and UDP as transport protocols.

INTRODUCTION

VoIP applications require low delay and jitter. UDP is preferred because it sends data continuously without retransmissions, while TCP introduces latency through acknowledgments.

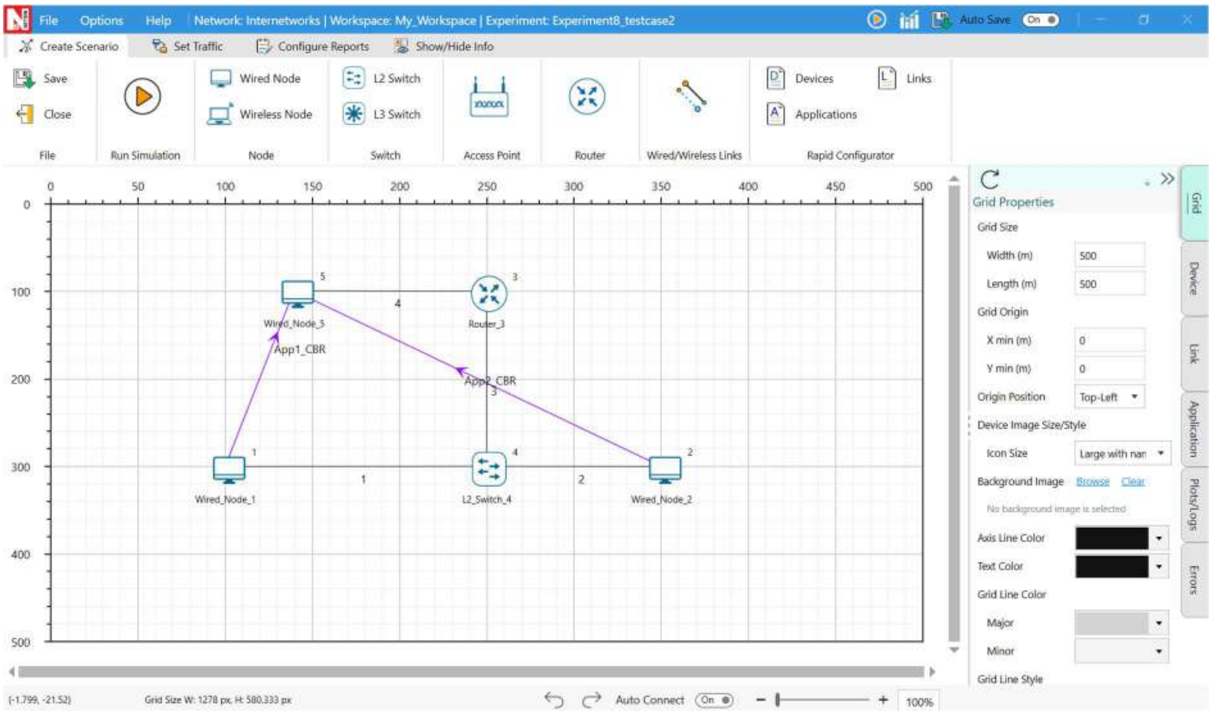
THEORY

VoIP streams are real-time and sensitive to delay. Retransmissions in TCP disrupt audio flow, whereas UDP allows minor packet loss without breaking continuity.

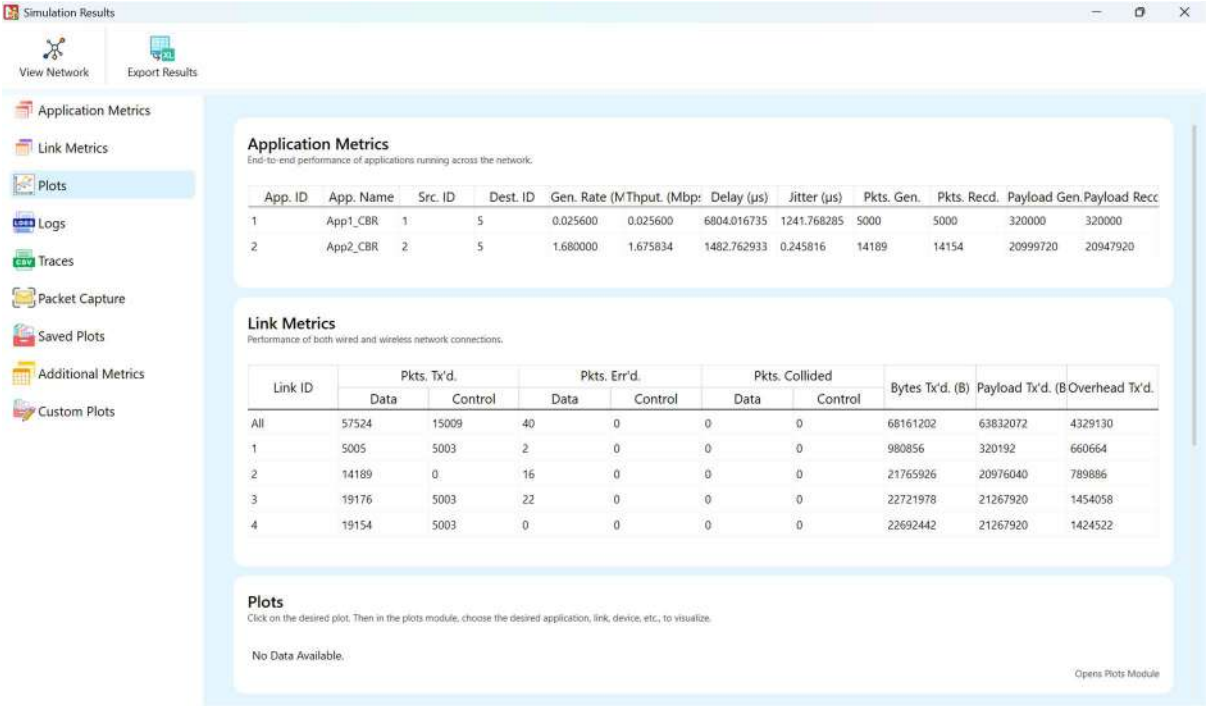
SETUP / PROCEDURE

1. Create a VoIP session using TCP and UDP.
2. Measure delay, jitter, and packet loss.
3. Compare the two protocols.

TOPOLOGY:



TESTCASE 1:



Simulation Results

View Network | Export Results

- Application Metrics
- Link Metrics
- Plots
- Logs
- Traces
- Packet Capture
- Saved Plots
- Additional Metrics
- Custom Plots

Application Metrics

End-to-end performance of applications running across the network.

App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_CBR	1	5	0.025600	0.025600	6804.016735	1241.768285	5000	5000	320000	320000
2	App2_CBR	2	5	1.680000	1.675834	1482.762933	0.245816	14189	14154	20999720	20947920

Link Metrics

Performance of both wired and wireless network connections.

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	57524	15009	40	0	0	0	68161202	63832072	4329130
1	5005	5003	2	0	0	0	980856	320192	660664
2	14189	0	16	0	0	0	21765926	20976040	789886
3	19176	5003	22	0	0	0	22721978	21267920	1454058
4	19154	5003	0	0	0	0	22692442	21267920	1424522

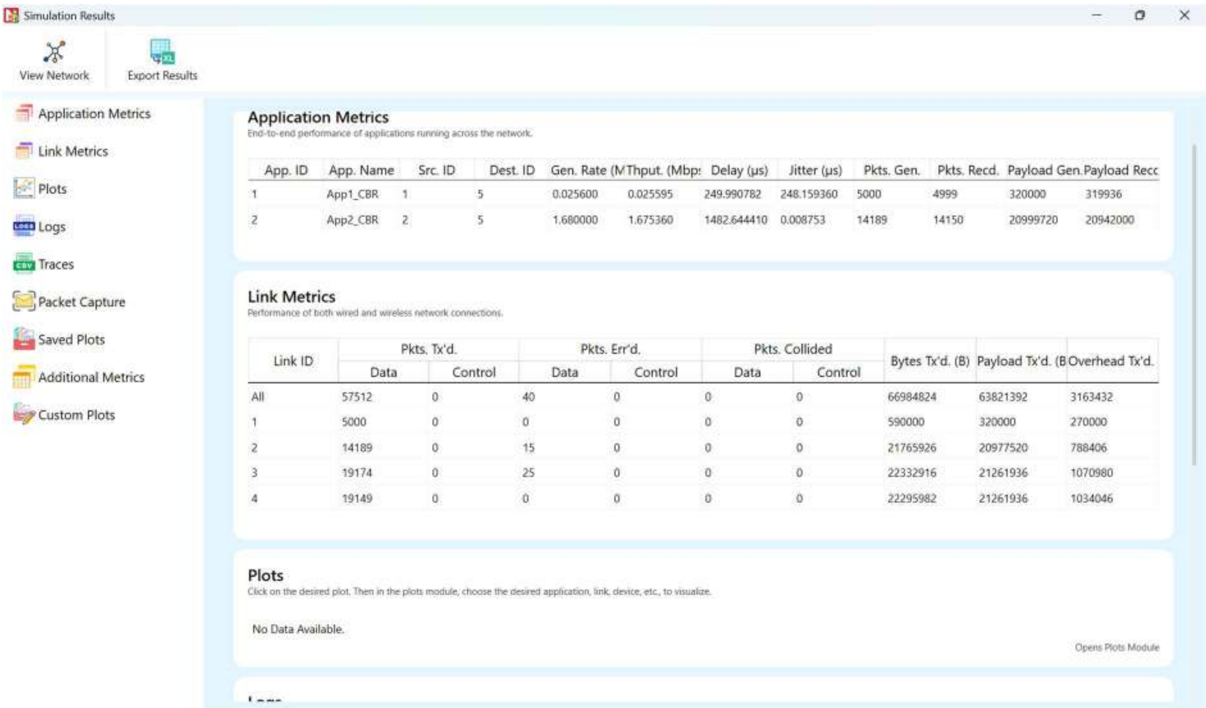
Plots

Click on the desired plot. Then in the plots module, choose the desired application, link, device, etc., to visualize.

No Data Available.

Opens Plots Module

TESTCASE 2:



Simulation Results

View Network | Export Results

- Application Metrics
- Link Metrics
- Plots
- Logs
- Traces
- Packet Capture
- Saved Plots
- Additional Metrics
- Custom Plots

Application Metrics

End-to-end performance of applications running across the network.

App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_CBR	1	5	0.025600	0.025595	249.990782	248.159360	5000	4999	320000	319936
2	App2_CBR	2	5	1.680000	1.675360	1482.644410	0.008753	14189	14150	20999720	20942000

Link Metrics

Performance of both wired and wireless network connections.

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	57512	0	40	0	0	0	66984824	63821392	3163432
1	5000	0	0	0	0	0	590000	320000	270000
2	14189	0	15	0	0	0	21765926	20977520	788406
3	19174	0	25	0	0	0	22332916	21261936	1070980
4	19149	0	0	0	0	0	22295982	21261936	1034046

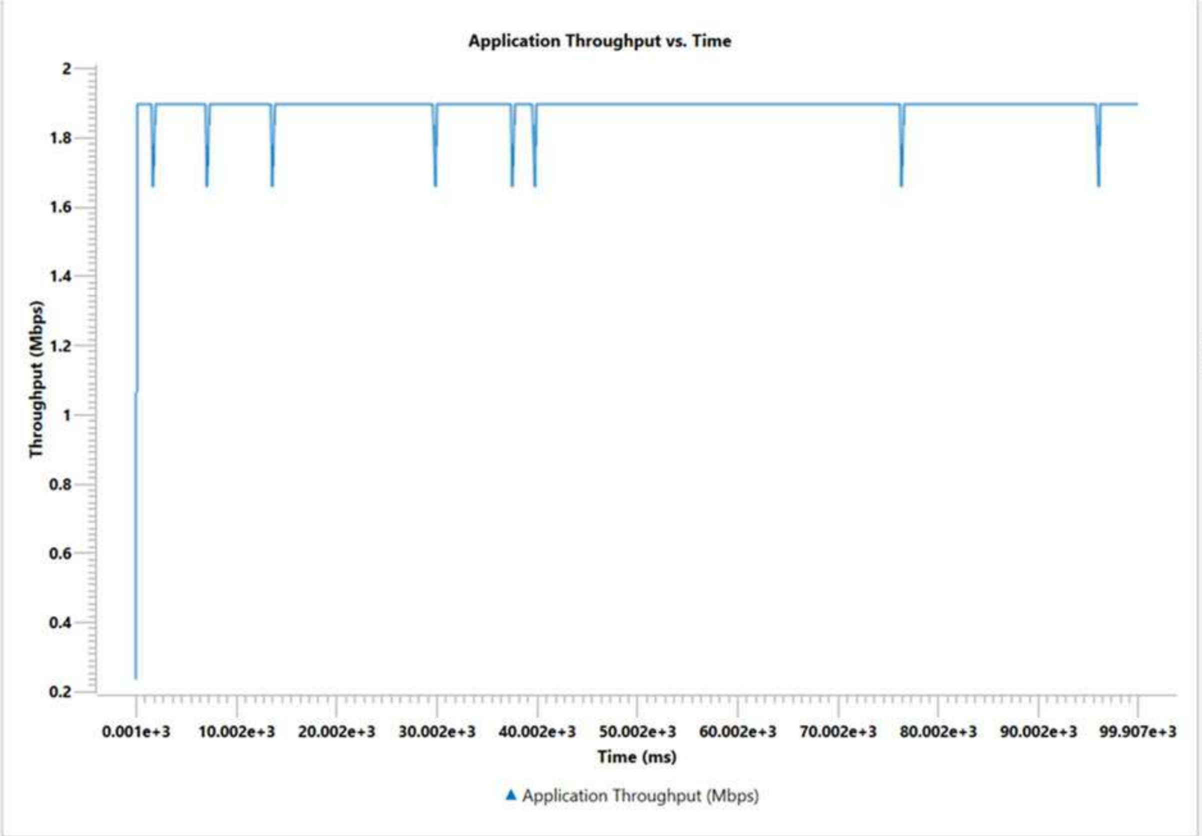
Plots

Click on the desired plot. Then in the plots module, choose the desired application, link, device, etc., to visualize.

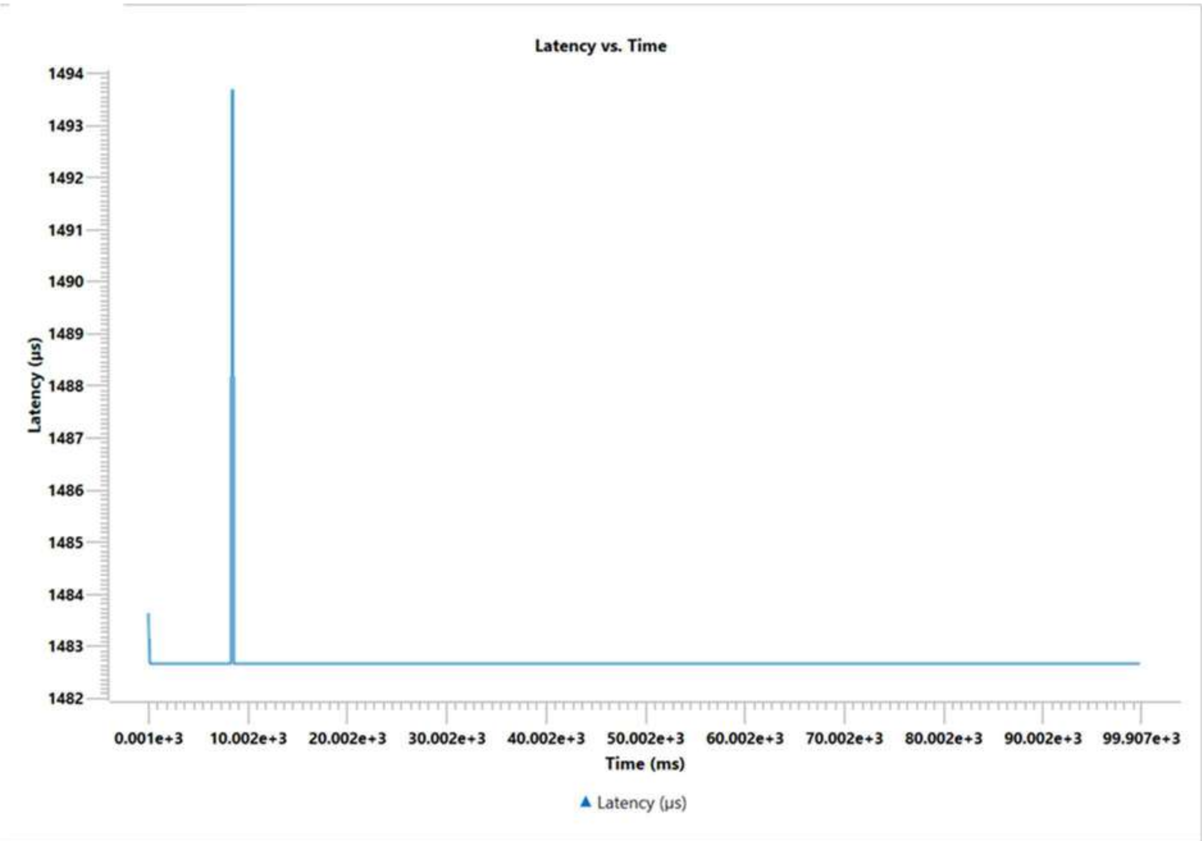
No Data Available.

Opens Plots Module

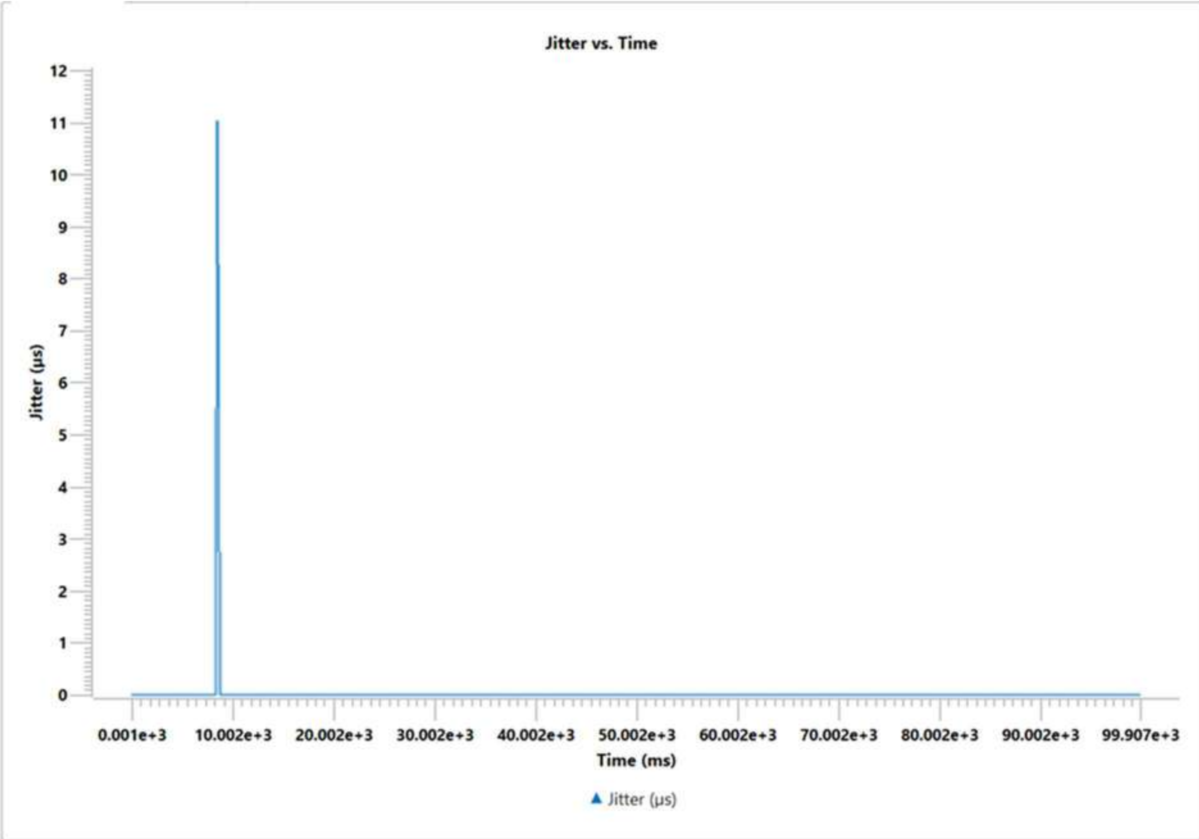
GRAPHS :



Application Throughput vs. Time
Throughput (Mbps)
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0.001e+3
10.002e+3
20.002e+3
30.002e+3
40.002e+3
50.002e+3
60.002e+3
70.002e+3
80.002e+3
90.002e+3
99.907e+3
Time (ms)
Application Throughput (Mbps)



Latency vs. Time
Latency (µs)
1494
1493
1492
1491
1490
1489
1488
1487
1486
1485
1484
1483
1482
0.001e+3
10.002e+3
20.002e+3
30.002e+3
40.002e+3
50.002e+3
60.002e+3
70.002e+3
80.002e+3
90.002e+3
99.907e+3
Time (ms)
Latency (µs)



Observation

UDP shows lower delay and jitter, while TCP introduces retransmission delays.

Conclusion

UDP is more suitable for VoIP due to lower latency, despite minor packet losses.

Experiment 8: Network Congestion Test

AIM

To observe how multiple competing protocols (TCP and UDP) share bandwidth and experience congestion.

INTRODUCTION

When several flows share a link, congestion may occur if offered load exceeds capacity. TCP reduces its sending rate, while UDP continues sending, possibly starving TCP flows.

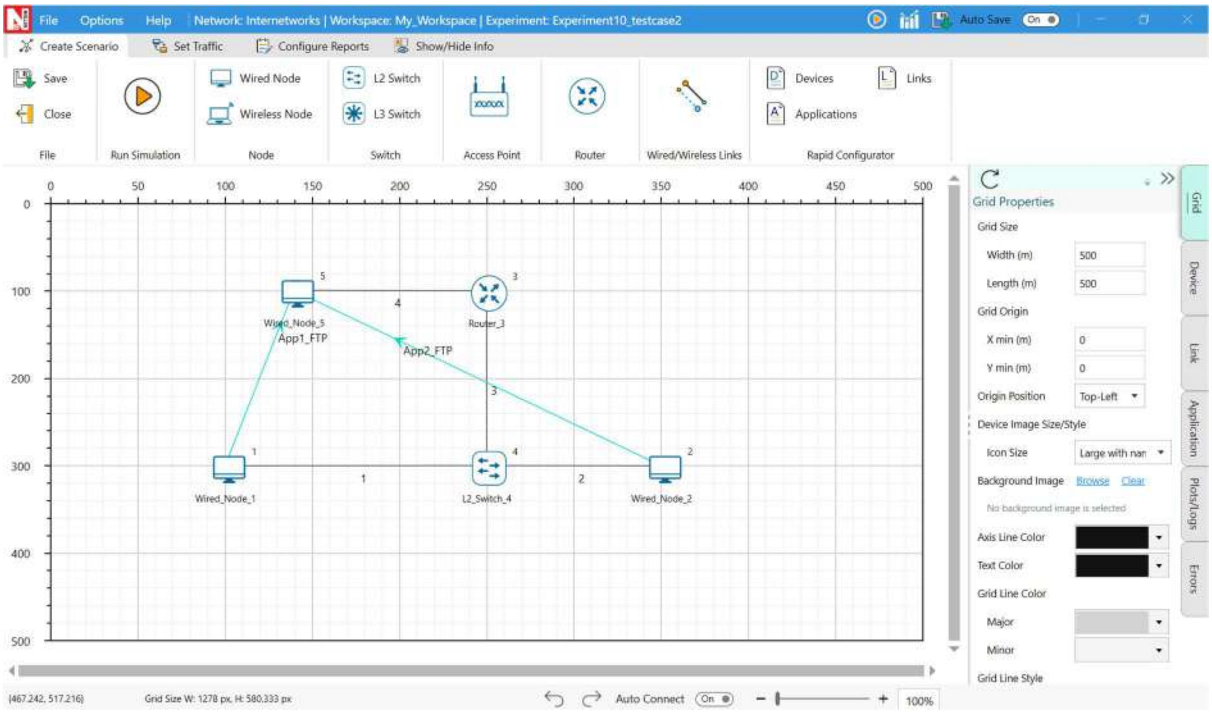
THEORY

TCP's congestion control adjusts window size on detecting packet loss. UDP, being uncontrolled, can dominate bandwidth, causing unfair throughput distribution.

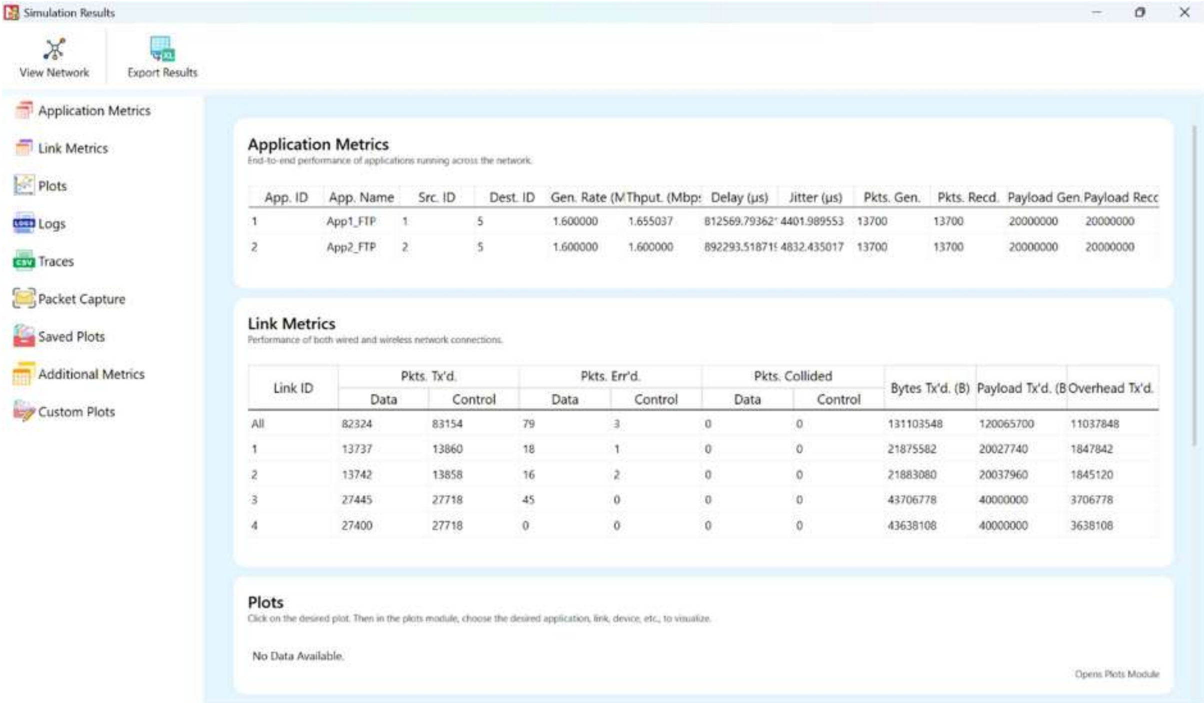
SETUP / PROCEDURE

1. Create a mixed traffic scenario (TCP + UDP).
2. Vary UDP background traffic rate.
3. Record throughput, fairness, and packet drops.

TOPOLOGY:



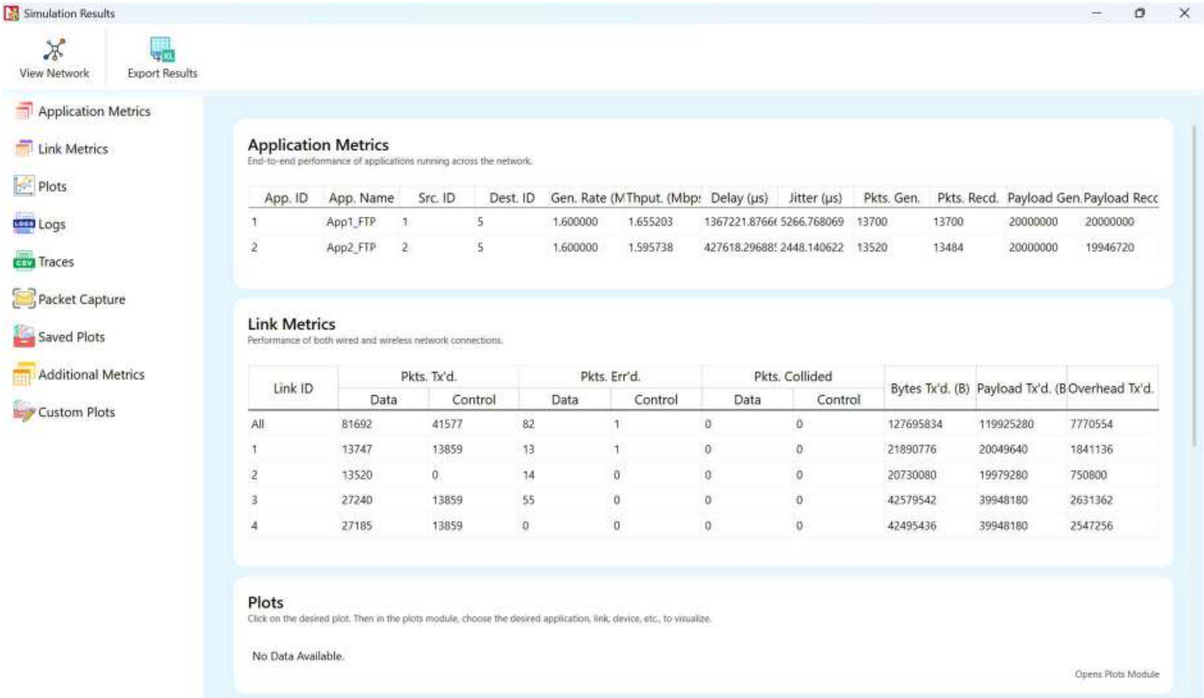
TESTCASE 1:



App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	5	1.600000	1.655037	812569.79362'	4401.989553	13700	13700	20000000	20000000
2	App2_FTP	2	5	1.600000	1.600000	892293.51871!	4832.435017	13700	13700	20000000	20000000

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	82324	83154	79	3	0	0	131103548	120065700	11037848
1	13737	13860	18	1	0	0	21875582	20027740	1847842
2	13742	13858	16	2	0	0	21883080	20037960	1845120
3	27445	27718	45	0	0	0	43706778	40000000	3706778
4	27400	27718	0	0	0	0	43638108	40000000	3638108

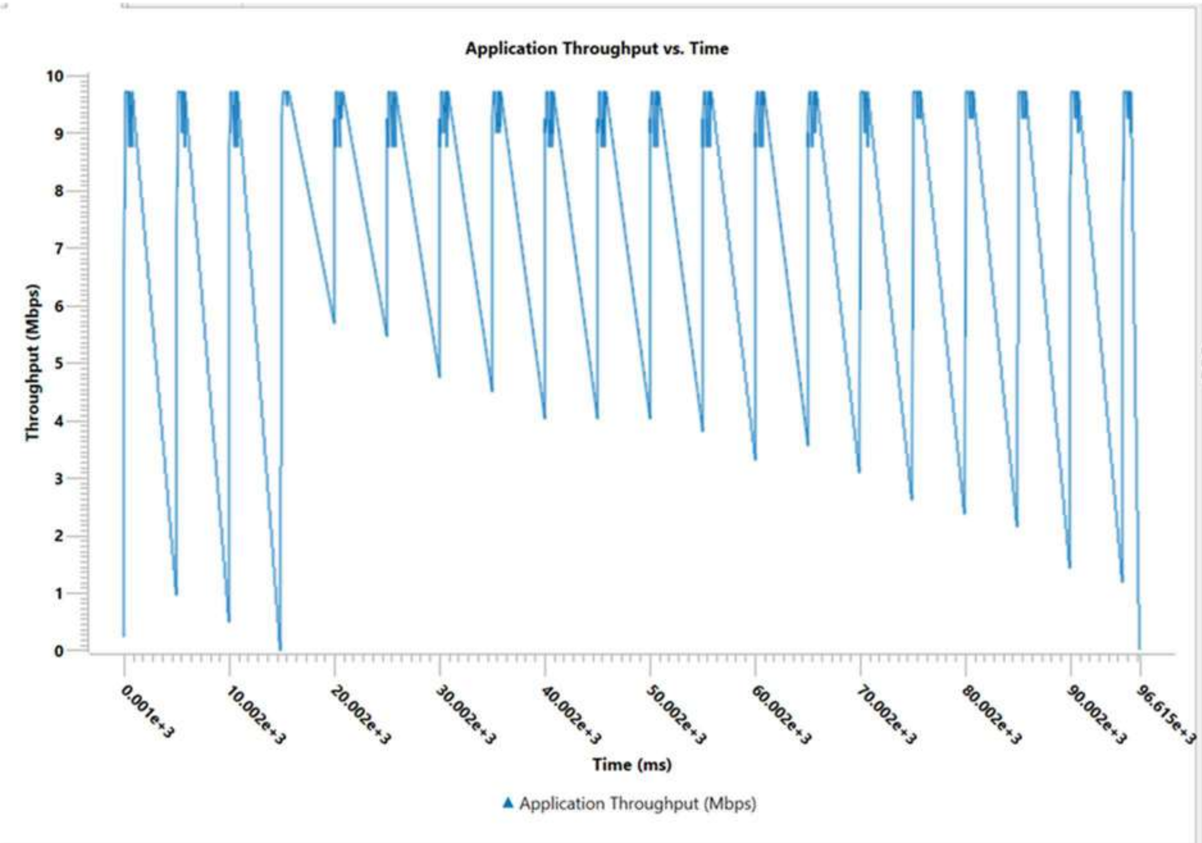
TESTCASE 2:

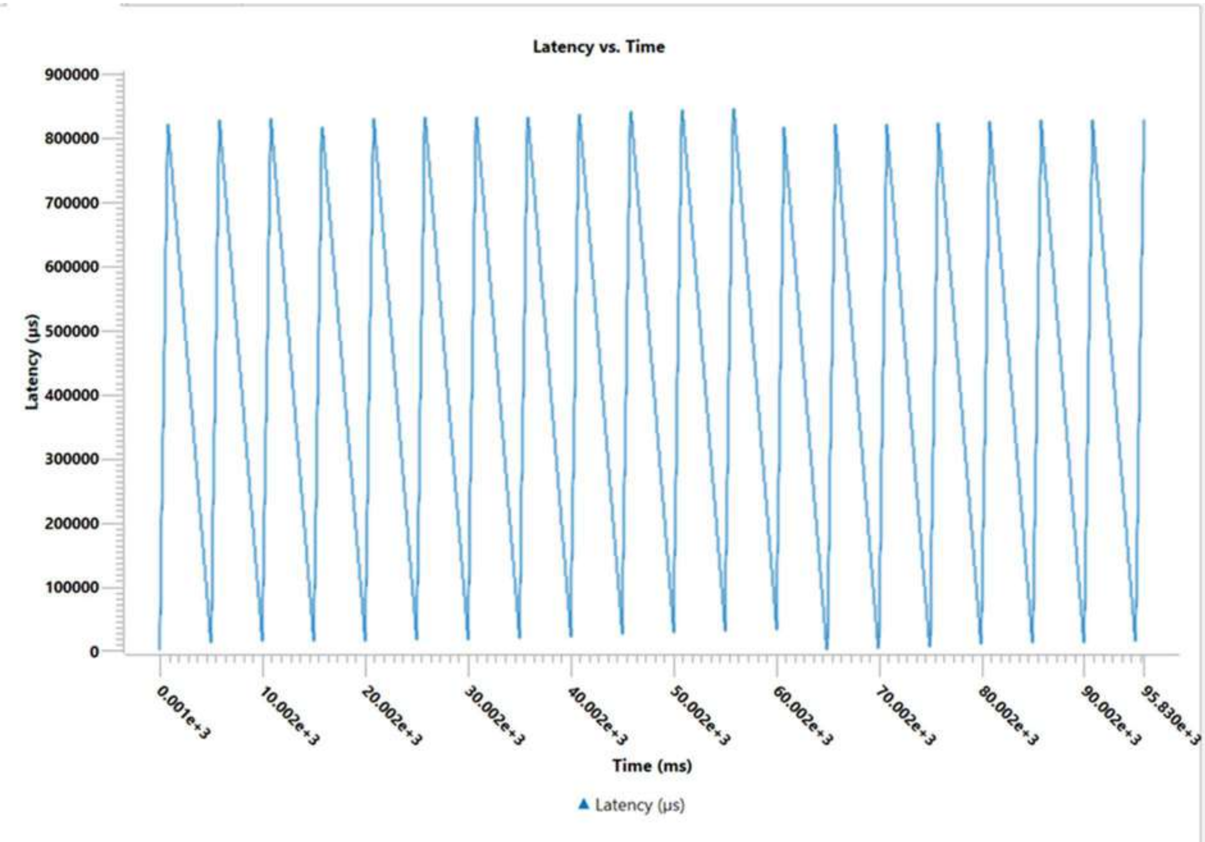


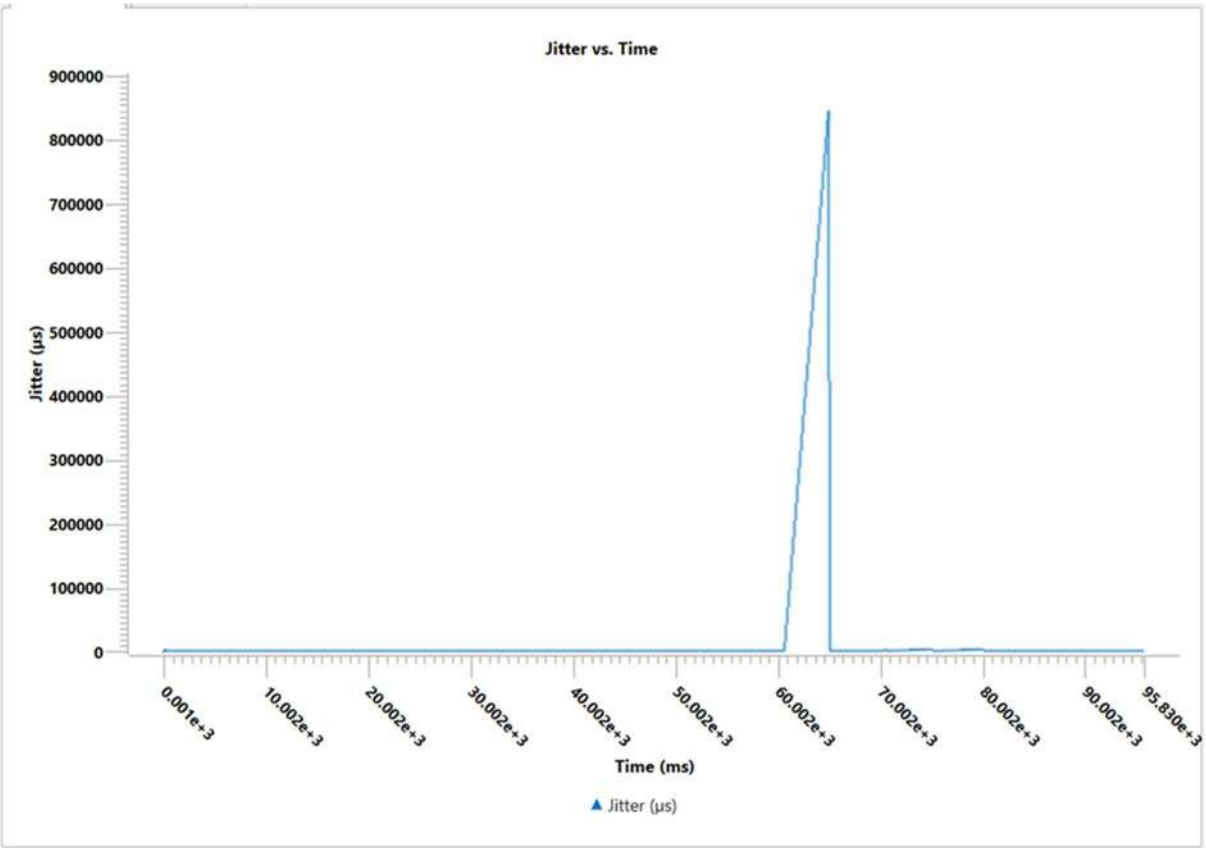
App. ID	App. Name	Src. ID	Dest. ID	Gen. Rate (M	Thput. (Mbp:	Delay (µs)	Jitter (µs)	Pkts. Gen.	Pkts. Recd.	Payload Gen.	Payload Recc
1	App1_FTP	1	5	1.600000	1.655203	1367221.8766	5266.768069	13700	13700	20000000	20000000
2	App2_FTP	2	5	1.600000	1.595738	427618.29688!	2448.140622	13520	13484	20000000	19946720

Link ID	Pkts. Tx'd.		Pkts. Err'd.		Pkts. Collided		Bytes Tx'd. (B)	Payload Tx'd. (B	Overhead Tx'd.
	Data	Control	Data	Control	Data	Control			
All	81692	41577	82	1	0	0	127695834	119925280	7770554
1	13747	13859	13	1	0	0	21890776	20049640	1841136
2	13520	0	14	0	0	0	20730080	19979280	750800
3	27240	13859	55	0	0	0	42579542	39948180	2631362
4	27185	13859	0	0	0	0	42495436	39948180	2547256

GRAPHS:


Application Throughput vs. Time
Throughput (Mbps)
10
9
8
7
6
5
4
3
2
1
0
0.001e+3
10.002e+3
20.002e+3
30.002e+3
40.002e+3
50.002e+3
60.002e+3
70.002e+3
80.002e+3
90.002e+3
96.615e+3
Time (ms)
Application Throughput (Mbps)


Latency vs. Time
Latency (µs)
900000
800000
700000
600000
500000
400000
300000
200000
100000
0
0.001e+3
10.002e+3
20.002e+3
30.002e+3
40.002e+3
50.002e+3
60.002e+3
70.002e+3
80.002e+3
90.002e+3
95.830e+3
Time (ms)
Latency (µs)



Observation

As UDP traffic increases, TCP throughput drops sharply due to congestion.

Conclusion

TCP and UDP compete for bandwidth; TCP adapts to congestion, while UDP can cause unfairness under heavy load.